MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

GATE, NAND, DIGITAL, KEY CODE LHH

This specification is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic NAND gate module.

1.2 Classification. This specification covers one class of modules.

Class I as specified in MIL-STD-1389.

2. APPLICABLE DOCUMENTS

* 2.1 Government specifications and standards. Unless otherwise specified, the following specifications and standards, of the issue listed in that issue of the Department of Defense Index of Specifications and Standards specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.2 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Naval Electronic Systems Command, ATTN: ELEX 8111, Department of the Navy, Washington, DC 20363, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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3.3 Failure rate requirements. All modules acquired to this specification shall have a predicted failure rate no greater than 0.38 failures per million operating hours when predicted in accordance with MIL-STD-1389.

3.4 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I as specified in MIL-STD-1389.

3.5 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.5.1 Key code. The module key code shall be LHH.

3.5.2 Size. The module size shall be 1A.

3.5.3 Weight. The module shall have a maximum weight of 35 grams.

3.5.4 Contact pins. The module shall have 40 contact pins positioned A1 through A40.

3.5.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.5.6 Depth. The module shall have no additional depth requirements.

3.5.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.5.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part number M28787/68-1, except no marking shall interfere with heat dissipation from the fin (for example, affixing a label which may be a heat barrier).

3.6 Circuit pin configuration. The function diagram and pin assignments shall conform to figure 1.

3.7 Acceptance tests. All production modules shall be tested in accordance with footnotes 1 and 2 of table I.

3.8 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits of table I (see 6.2).

3.9 Power supply voltage. The module shall be capable of conforming to the requirements of table I for a power supply voltage (V_{CC}) of 5.0 ±0.5 V dc, except that the limits for output short circuit current apply only for a power supply voltage of 5.5 V dc ±1 percent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

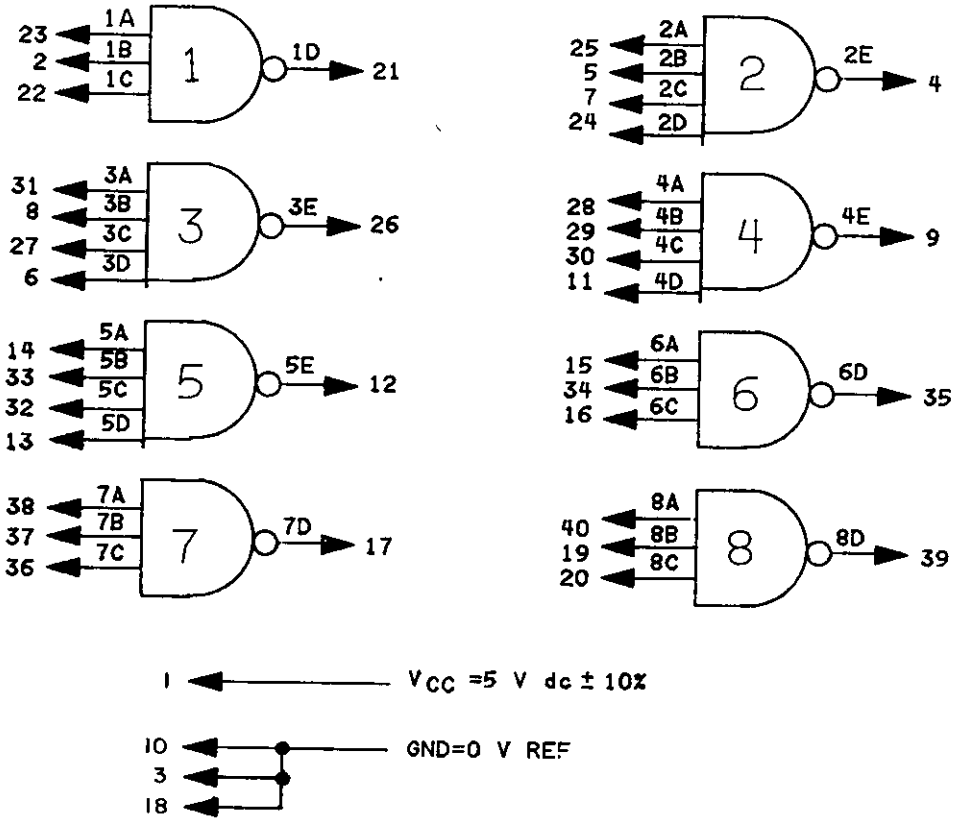
4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

* 4.3 Test equipment. Test equipment specified on figures and in tables with a type designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. All resistors shall be metal film and all capacitors shall be ceramic unless otherwise specified.

4.4 Test procedures. The module shall be tested as specified in table I.

4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Program Quality Assurance Activity and they conform to the requirements of MIL-M-28787.

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NOTES:
1. 1A-8A = Gate inputs.
2. 1B-8B = Gate inputs.
3. 1C-8C = Gate inputs.
4. 2D-5D = Gate inputs.
5. 1D, 6D, 7D, 8D = Gate outputs.
6. 2E-5E = Gate outputs.

*FIGURE 1. LHH module function diagram, NAND gate.

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*TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation	III	S	10	∞	1	∞	10	∞	1	∞	MΩ
B1-B8	V_{OUT} (low loaded)	IV	S	10	1000	10	1000	10	1000	10	1000	mV
C1,C22, C23,C28	Input breakdown current	V	S	0	2	0	2	0	2	0	2	mA
C2-C21, C24-C27	Input breakdown current	V	S	0	1	0	1	0	1	0	1	mA
D1-D8	Output short circuit current	VI	X	-40	-100	-40	-100	-40	-100	-40	-100	mA
E1-E8	Output leakage current	VII	X	0	1	0	1	0	1	0	1	mA
F1-F28	Input clamp diode voltage	VIII	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
G1-G8	Output voltage, low level	IX	X	0	500	0	500	0	500	0	500	mV
H1-H28	Output voltage, high level	X	X	2.5	4.5	2.5	4.5	2.5	4.5	2.5	4.5	V
J1,J22, J23,J28	Input current, high level	XI	X	0	100	0	100	0	100	0	100	μA
J2-J21, J24-J27	Input current, high level	XI	X	0	50	0	50	0	50	0	50	μA
K1-K28	Input current, low level	XII	X	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	-0.4	-2.2	mA
L1,L2	Continuity	XIII	X	0	1	0	1	0	1	0	1	Ω
L3,L4	Continuity	XIII	Q	0.0	0.1	0.0	0.1	0.0	0.1	0.0	0.1	Ω
M1	Power supply current, low level outputs	XIV	S	30	72	30	72	30	72	30	72	mA
M2	Power supply current, high level outputs	XIV	S	15	32	15	32	15	32	15	32	mA
N1-N8	Transition time (t_{TLH}), C_L = 50 pF	XV	S	2	15	2	15	2	18	2	18	ns

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
N9-N16	Transition time (t_{TLH}), C_L = 150 pF	XV	Q	5	27	5	27	5	30	5	30	ns
P1-P8	Transition time (t_{THL}), C_L = 50 pF	XV	S	2	6	2	6	2	8	2	8	ns
P9-P16	Transition time (t_{THL}), C_L = 150 pF	XV	Q	3	11	3	11	3	13	3	13	ns
Q1-Q28	Propagation delay time (t_{PLH}), C_L = 50 pF	XV	S	2	7	2	7	2	8	2	8	ns
Q29-Q56	Propagation delay time (t_{PLH}), C_L = 150 pF	XV	Q	3.0	10.5	3.0	10.5	3	12	3	12	ns
R1-R28	Propagation delay time (t_{PHL}), C_L = 50 pF	XV	S	2.0	7.5	2.0	7.5	2	8	2	8	ns
R29-R56	Propagation delay time (t_{PHL}), C_L = 150 pF	XV	Q	3.0	11.5	3.0	11.5	3	12	3	12	ns
S1-S4	Output to output delta propagation delay time, (Δt_{PLH}), C_L = 50 pF	XVI	S	-3	3	-3	3	-3	3	-3	3	ns
T1-T4	Output to output delta propagation delay time, (Δt_{PHL}), C_L = 50 pF	XVI	S	-3	3	-3	3	-3	3	-3	3	ns
U1-U8	Output crosstalk, low level	XVII	Q	0	500	0	500	0	500	0	500	mV
V1-V8	Output crosstalk, high level	XVII	Q	0	400	0	400	0	400	0	400	mV
W	Power supply transient amplitude	XVIII	Q	0	400	0	400	0	400	0	400	mV
X	Power supply filter capacitance	XIX	S	1.6	3.0	1.6	3.0	1.6	3.0	1.6	3.0	µF

See footnotes at end of table.

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*TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	Acceptance requirements 25°C ±5°C ambient temperature				Operating temperature 0°C to 60°C				Unit
				Initial		End-of-life		Initial		End-of-life		
				Min	Max	Min	Max	Min	Max	Min	Max	
Y1,Y22, Y23,Y28	Input capacitance	XX	Q	5	30	5	30	5	30	5	30	pF
Y2-Y21, Y24-Y27	Input capacitance	XX	Q	5	15	5	15	5	15	5	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.

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4.4.2 Order of testing. Test conditions A through Y shall be performed in any order with the following exceptions. If test conditions A, B, C, D, or E are part of a sequence of tests, the order of testing shall be as follows: A, B, C, D, E where applicable, followed by the remaining tests in the sequence.

4.5 Qualification testing. Initial qualification tests and periodic-check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through Y in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ±5°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions D through X shall be performed in the order specified in 4.4.2 using the initial limits for applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 2.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests of table I which are marked with an "X" in the test type column, and test condition X, shall be performed using the initial limits at 25°C ±5°C ambient.

4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 2. Following the life test and while still at the maximum class temperature, electrical tests D through X shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements of table I. Following the return of the module to 25°C, electrical tests D through Y shall be performed in the order specified in 4.4.2 using the 25°C ±5°C ambient end-of-life requirements of table I.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

6. NOTES

6.1 Definitions. Definitions are as specified in MIL-STD-1331.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements of table I, the module is considered to have completed its useful life (see 3.8).

6.3 Cross reference of classification. Part number M28787/68-1 supersedes part number M28787/68. The two parts are interchangeable.

* 6.4 Module description. The following electrical parameters reflect the 0°C to 60°C end-of-life limits from table I.

Maximum power dissipation (V_{CC} = 5.5 V dc)	
All outputs at low level	396 mW (outputs not loaded)
All outputs at high level	176 mW (outputs not loaded)
Supply voltage	5 V dc ±10 percent
Input voltage	
Low level	-0.5 minimum, 0.8 maximum
High level	2 V minimum, V_{CC} maximum

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Input current, low level	
V_{IL} = 0.0 V	-2.2 mA maximum
V_{IL} = 0.5 V	-2 mA maximum
Input current, high level	
Inputs 1A, 7A, 6C, and 8C	
V_{IH} = 2.5 V	100 µA maximum
V_{IH} = 5.5 V	2 mA maximum
All inputs except 1A, 7A, 6C and 8C	
V_{IH} = 2.5 V	50 µA maximum
V_{IH} = 5.5 V	1 mA maximum
Output voltage	
Low level (I_{OL} = 20 mA)	0.5 V maximum
High level (I_{OH} = -1 mA)	2.5 V minimum
Output drive current maximum capability	
Low level (V_{OL} = 0.5 V)	20 mA
High level (V_{OH} = 2.5 V)	-1 mA
Propagation delay time (t_{PHL})	
50 pF load	2 ns minimum, 8 ns maximum
150 pF load	3 ns minimum, 12 ns maximum
Propagation delay time (t_{PLH})	
50 pF load	2 ns minimum, 8 ns maximum
150 pF load	3 ns minimum, 12 ns maximum
Delta propagation delay time	
Δt_{PHL} (50 pF load)	
Between gates 1 and 2	3 ns maximum
Between gates 3 and 4	3 ns maximum
Between gates 5 and 6	3 ns maximum
Between gates 7 and 8	3 ns maximum
Delta propagation delay time	
Δt_{PLH} (50 pF load)	
Between gates 1 and 2	3 ns maximum
Between gates 3 and 4	3 ns maximum
Between gates 5 and 6	3 ns maximum
Between gates 7 and 8	3 ns maximum

* 6.5 Module function. This module contains four independent 3-input NAND gates and four independent 4-input NAND gates.

6.5.1 Input to output relationship. Each NAND gate has the input to output relationship of table II for positive logic.

6.5.2 Parallel outputs. Only the following outputs may be paralleled provided their corresponding inputs are parallel:

a. Outputs from gates 1 and 2 may be paralleled.
b. Outputs from gates 3 and 4 may be paralleled.
c. Outputs from gates 5 and 6 may be paralleled.
d. Outputs from gates 7 and 8 may be paralleled.

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TABLE II. Input to output relationship. 1/ 2/

3-input NAND GATE				4-input NAND gate				
Inputs			Output	Inputs				Output
A	B	C	D	A	B	C	D	E
L	L	L	H	L	L	L	L	H
L	L	H	H	L	L	L	H	H
L	H	L	H	L	L	H	L	H
L	H	H	H	L	L	H	H	H
H	L	L	H	L	H	L	L	H
H	L	H	H	L	H	L	H	H
H	H	L	H	L	H	H	L	H
H	H	H	L	L	H	H	H	H
				H	L	L	L	H
				H	L	L	H	H
				H	L	H	L	H
				H	L	H	H	H
				H	H	L	L	H
				H	H	L	H	H
				H	H	H	L	H
				H	H	H	H	L

1/ H = High level.
2/ L = Low level.

6.5.3 Ground pins. All electrical parameters are measured with pins 3, 10, and 18 connected separately to the 0 volt return. System problems may result from a different grounding method.

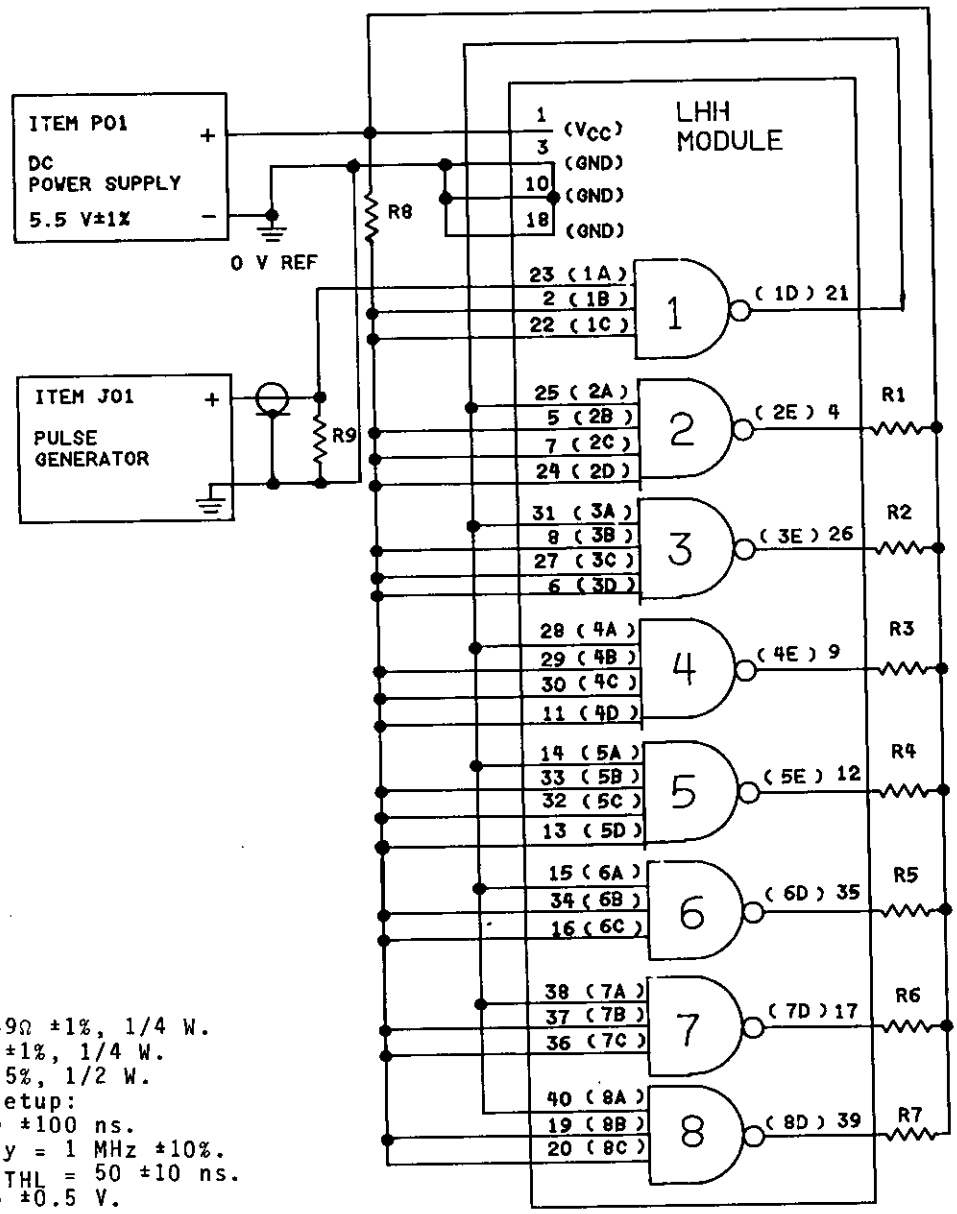
6.6 Changes from previous issue. The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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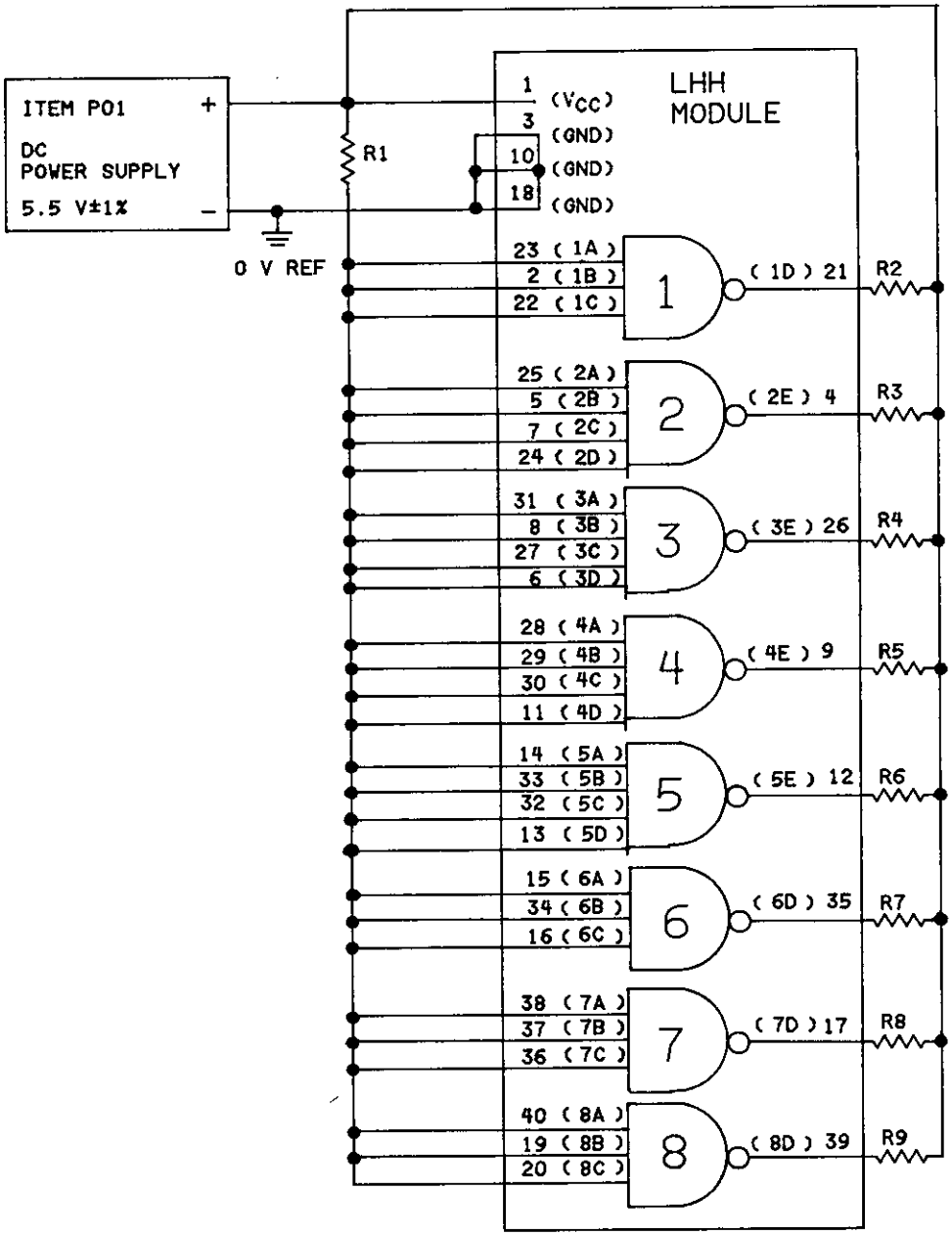
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NOTES:
1. R1-R7 = 249Ω ±1%, 1/4 W.
2. R8 = 1 kΩ ±1%, 1/4 W.
3. R9 = 51Ω ±5%, 1/2 W.
4. Item J01 setup:
   PW = 500 ±100 ns.
   Frequency = 1 MHz ±10%.
   $t_{TLH} = t_{THL}$ = 50 ±10 ns.
   PA = 3.5 ±0.5 V.

*FIGURE 2. Operating temperature and life test circuit.

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NOTES:
1. R1 = 1 kΩ ±1%, 1/4 W.
2. R2-R9 = 249Ω ±1%, 1/4 W.

FIGURE 3. Component temperature test circuit.

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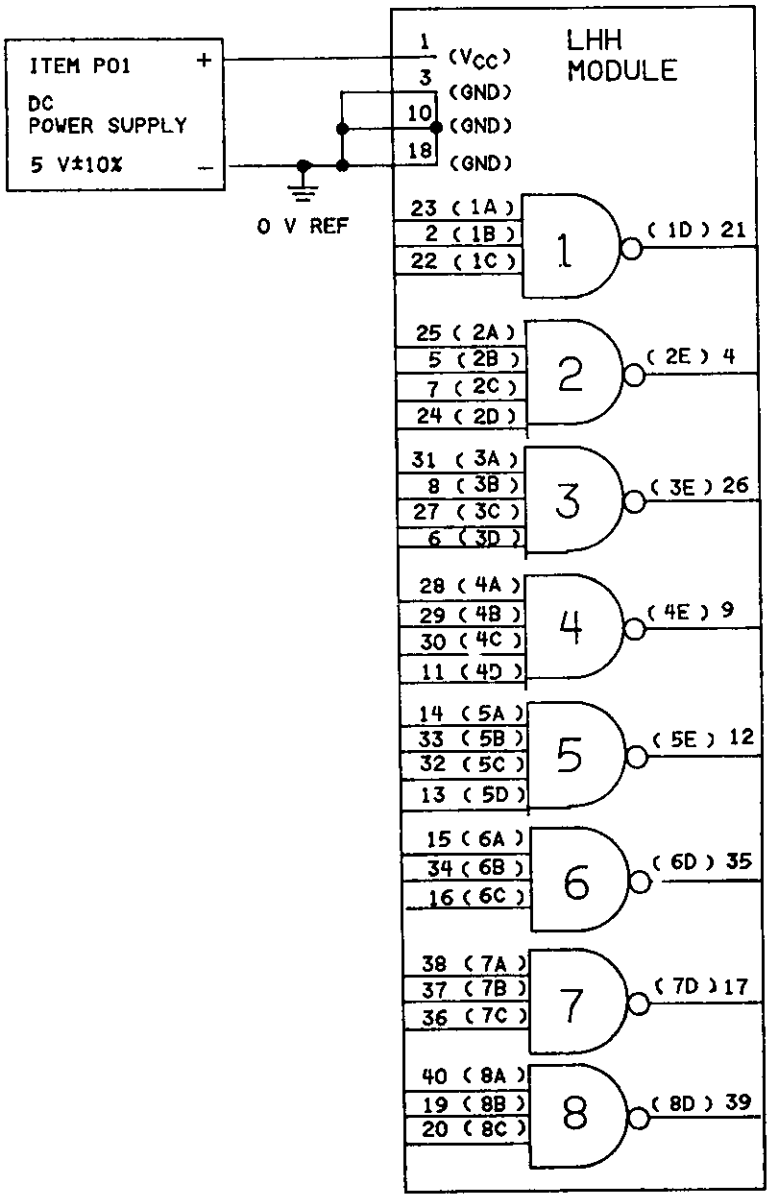


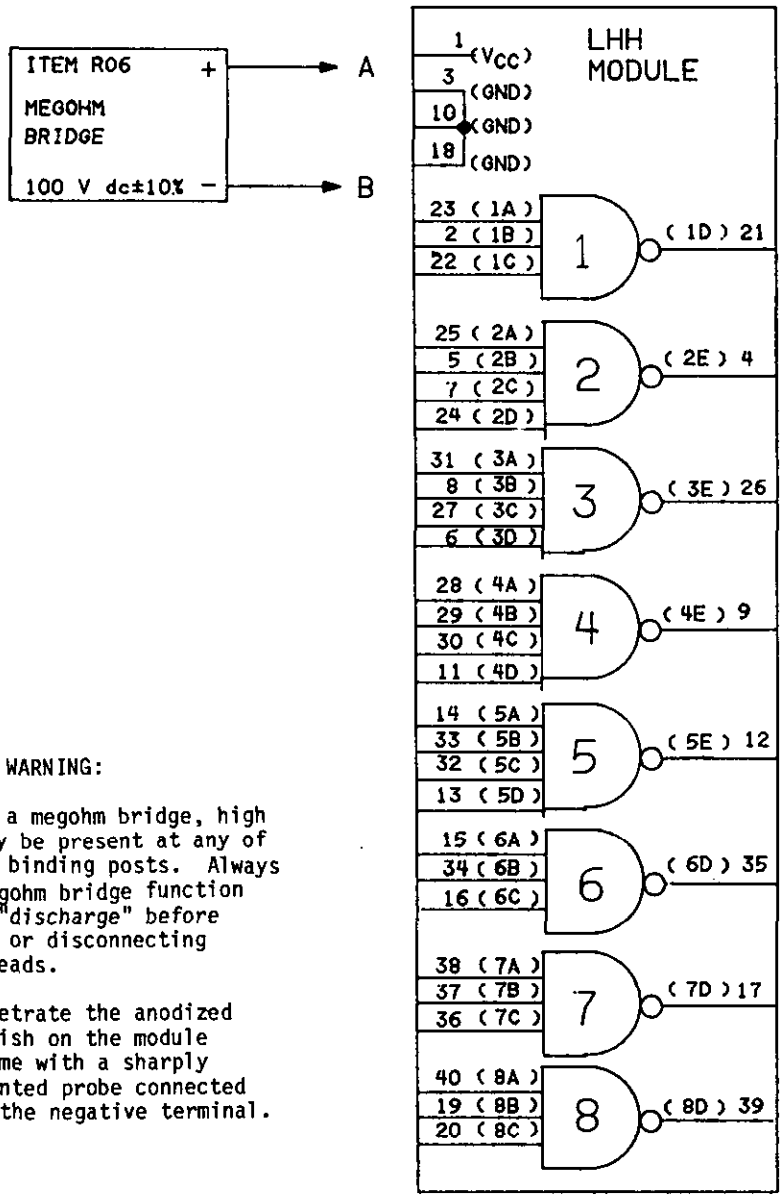
FIGURE 4. Vibration test circuit.

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TABLE III. Isolation test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A	Isolation	All pins	Frame	10	∞	MΩ	5

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WARNING:

When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "discharge" before connecting or disconnecting the test leads.

NOTE: Penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal.

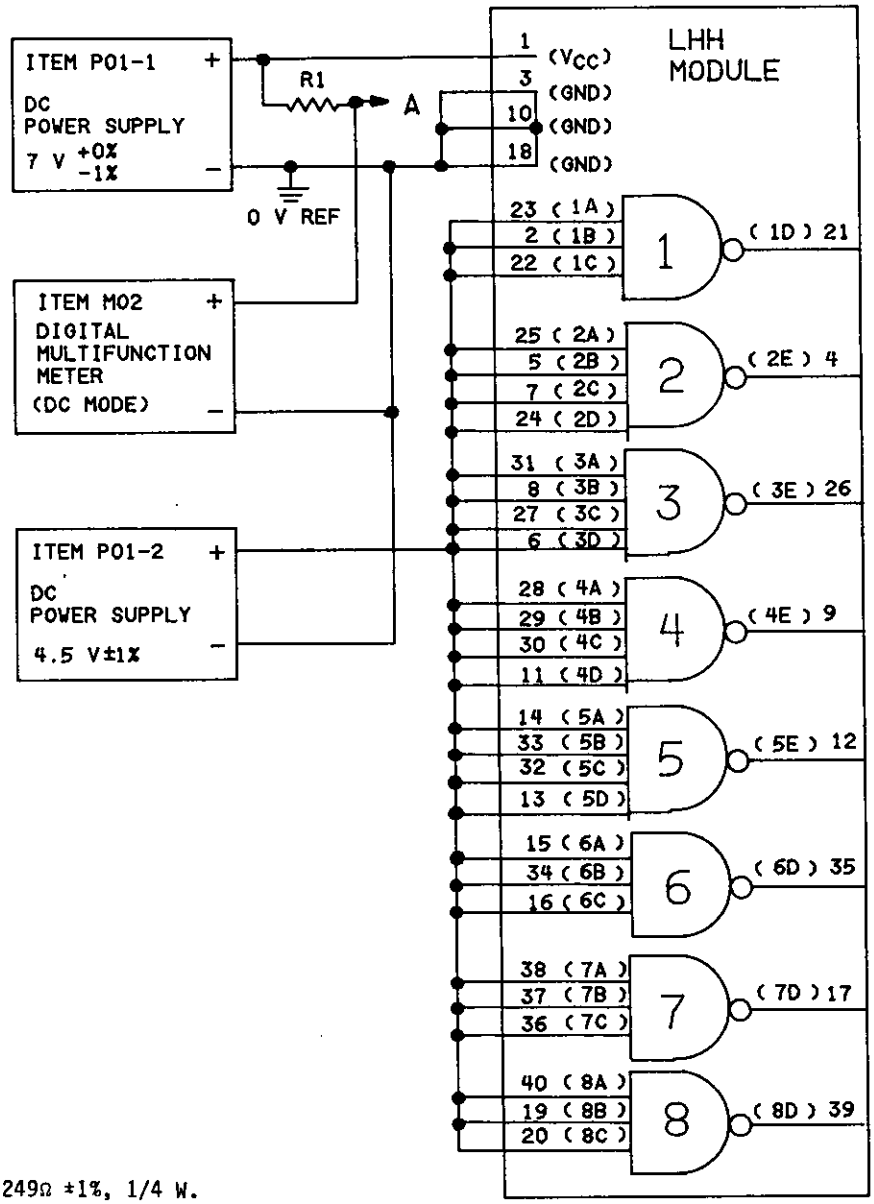
FIGURE 5. Isolation test circuit for table III.

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*TABLE IV. V_{OUT} (low-loaded) test.

Test condition	Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient. Min	Max	Unit	Test figure
B1	1D V_{OUT} (low loaded)	21	10	1000	mV	6
B2	2E "	4	"	"	"	"
B3	3E "	26	"	"	"	"
B4	4E "	9	"	"	"	"
B5	5E "	12	"	"	"	"
B6	6D "	35	"	"	"	"
B7	7D "	17	"	"	"	"
B8	8D "	39	"	"	"	"

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NOTE: R1 = 249Ω ±1%, 1/4 W.

*FIGURE 6. V_{OUT} (low-loaded) test circuit for table IV.

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TABLE V. Input breakdown current test.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C1	1A	Input breakdown current	23	2,5,6,7,8,11,13,14,15,16, 19,20,22,24,25,27,28,29,30, 31,32,33,34,36,37,38,40	0	2	mA	7
C2	1B	"	2	5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	1	"	"
C3	1C	"	22	2,5,6,7,8,11,13,14,15,16, 19,20,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40	"	"	"	"
C4	2A	"	25	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,27,28,29,30, 31,32,33,34,36,37,38,40	"	"	"	"
C5	2B	"	5	2,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
C6	2C	"	7	2,5,6,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
C7	2D	"	24	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,25,27,28,29,30, 31,32,33,34,36,37,38,40	"	"	"	"
C8	3A	"	31	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,32,33,34,36,37,38,40	"	"	"	"
C9	3B	"	8	2,5,6,7,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
C10	3C	"	27	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,28,29,30, 31,32,33,34,36,37,38,40	"	"	"	"
C11	3D	"	6	2,5,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

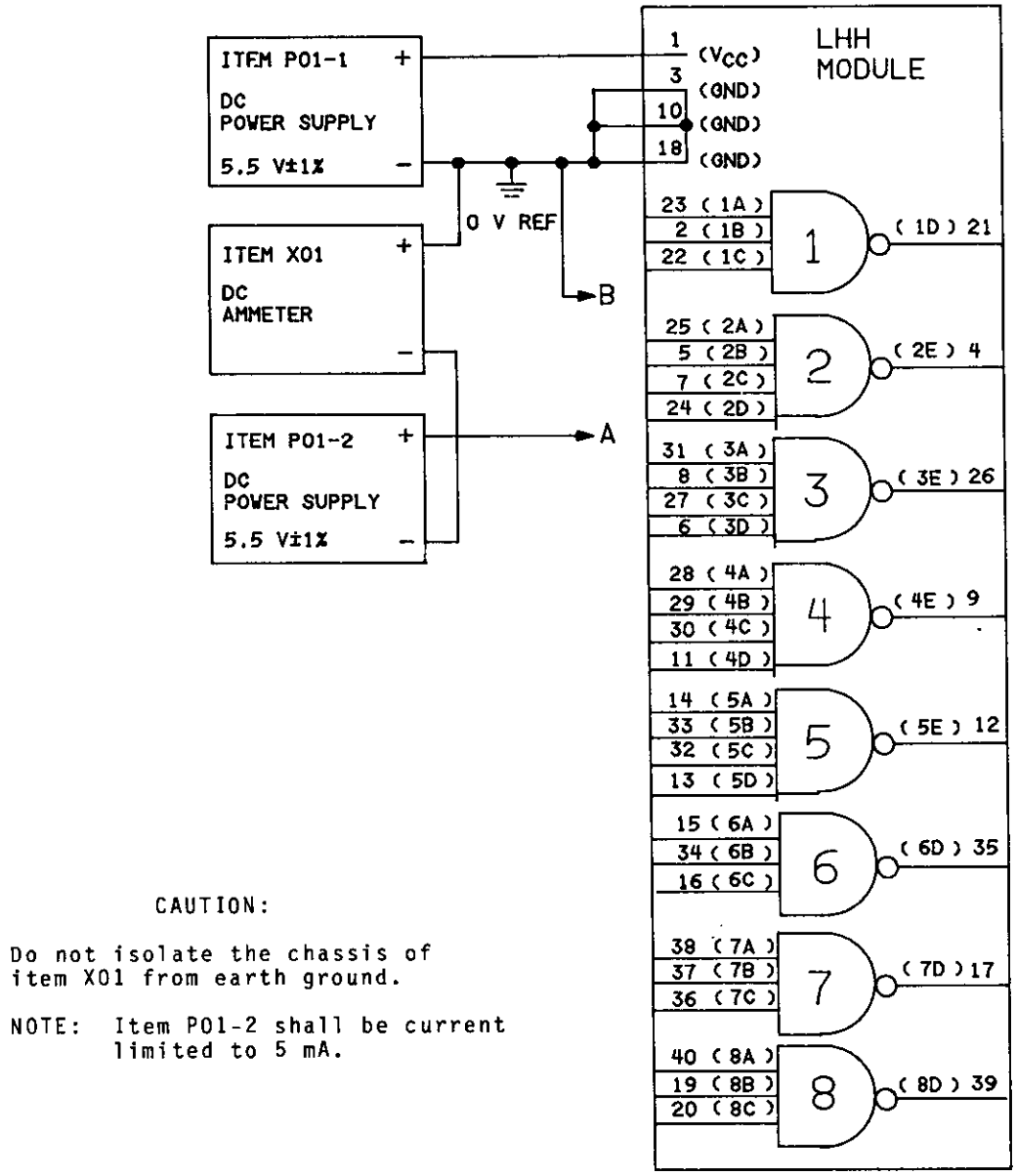
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
C12	4A Input breakdown current	28	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,29,30, 31,32,33,34,36,37,38,40	0	1	mA	7
C13	4B "	29	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,30, 31,32,33,34,36,37,38,40	"	"	"	"
C14	4C "	30	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 31,32,33,34,36,37,38,40	"	"	"	"
C15	4D "	11	2,5,6,7,8,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
C16	5A "	14	2,5,6,7,8,11,13,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
C17	5B "	33	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,34,36,37,38,40	"	"	"	"
C18	5C "	32	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,33,34,36,37,38,40	"	"	"	"
C19	5D "	13	2,5,6,7,8,11,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
C20	6A "	15	2,5,6,7,8,11,13,14,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
C21	6B "	34	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,36,37,38,40	"	"	"	"

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TABLE V. Input breakdown current test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
C22	6C	Input breakdown current	16	2,5,6,7,8,11,13,14,15, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	0	2	mA	7
C23	7A	"	38	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,40	"	2	"	"
C24	7B	"	37	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,38,40	"	1	"	"
C25	7C	"	36	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,37,38,40	"	"	"	"
C26	8A	"	40	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38	"	"	"	"
C27	8B	"	19	2,5,6,7,8,11,13,14,15,16, 20,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40	"	"	"	"
C28	8C	"	20	2,5,6,7,8,11,13,14,15,16, 19,22,23,24,25,27,28,29,30, 31,32,33,34,36,37,38,40	"	2	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: Item P01-2 shall be current limited to 5 mA.

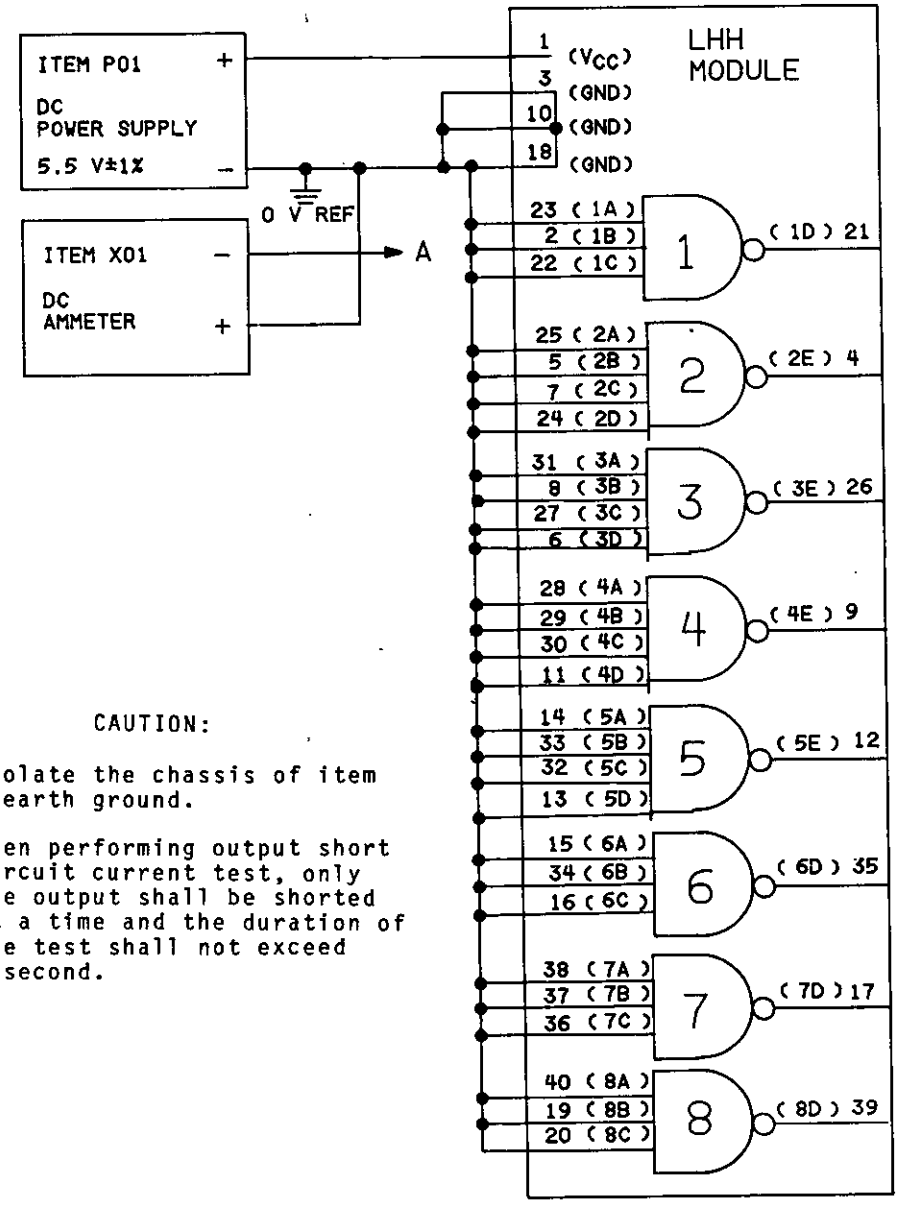
FIGURE 7. Input breakdown current test circuit for table V.

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TABLE VI. Output short circuit current test.

Test condition		Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
D1	1D	Output short circuit current	21	-40	-100	mA	8
D2	2E	"	4	"	"	"	"
D3	3E	"	26	"	"	"	"
D4	4E	"	9	"	"	"	"
D5	5E	"	12	"	"	"	"
D6	6D	"	35	"	"	"	"
D7	7D	"	17	"	"	"	"
D8	8D	"	39	"	"	"	"

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CAUTION:

Do not isolate the chassis of item XO1 from earth ground.

NOTE: When performing output short circuit current test, only one output shall be shorted at a time and the duration of the test shall not exceed 1 second.

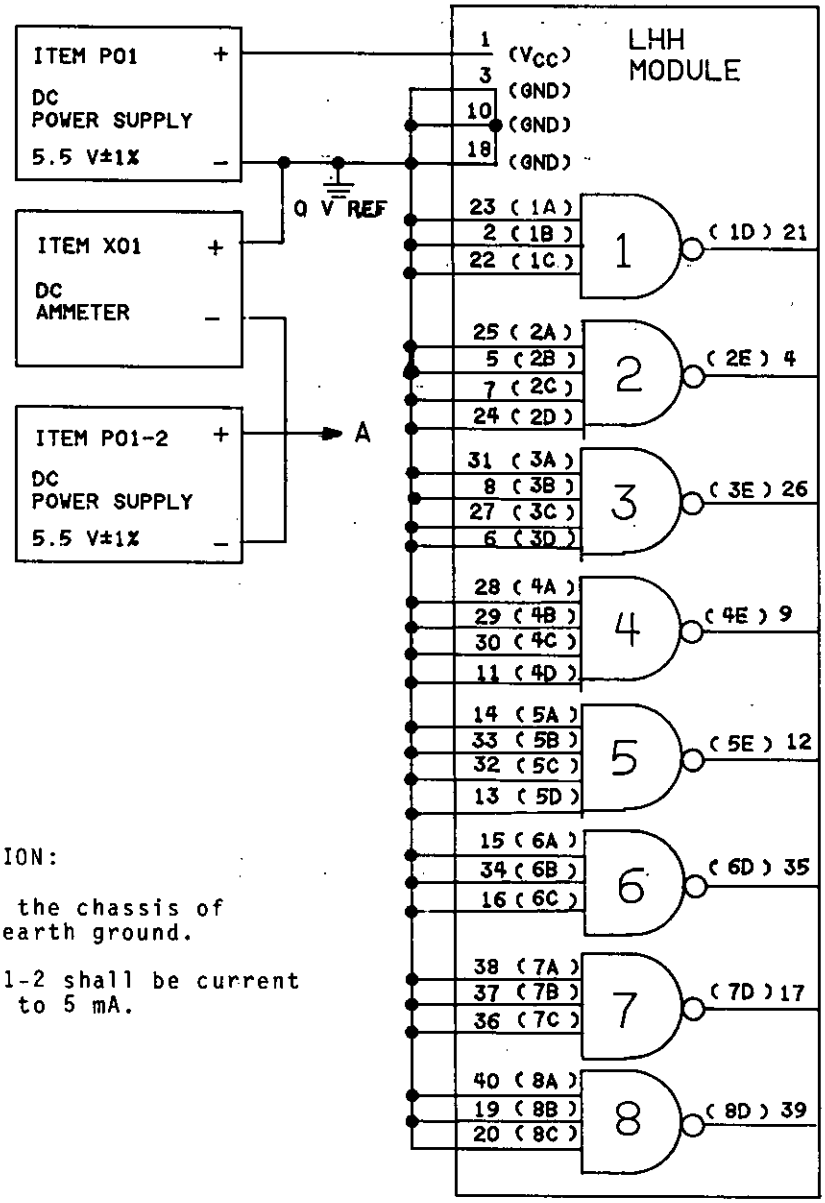
FIGURE 8. Output short circuit current test circuit for table VI.

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TABLE VII. Output leakage current test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
E1	1D Output leakage current	21	0	1	mA	9
E2	2E "	4	"	"	"	"
E3	3E "	26	"	"	"	"
E4	4E "	9	"	"	"	"
E5	5E "	12	"	"	"	"
E6	6D "	35	"	"	"	"
E7	7D "	17	"	"	"	"
E8	8D "	39	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

NOTE: Item P01-2 shall be current limited to 5 mA.

FIGURE 9. Output leakage current test circuit for table VII.

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*TABLE VIII. Input clamp diode voltage test.

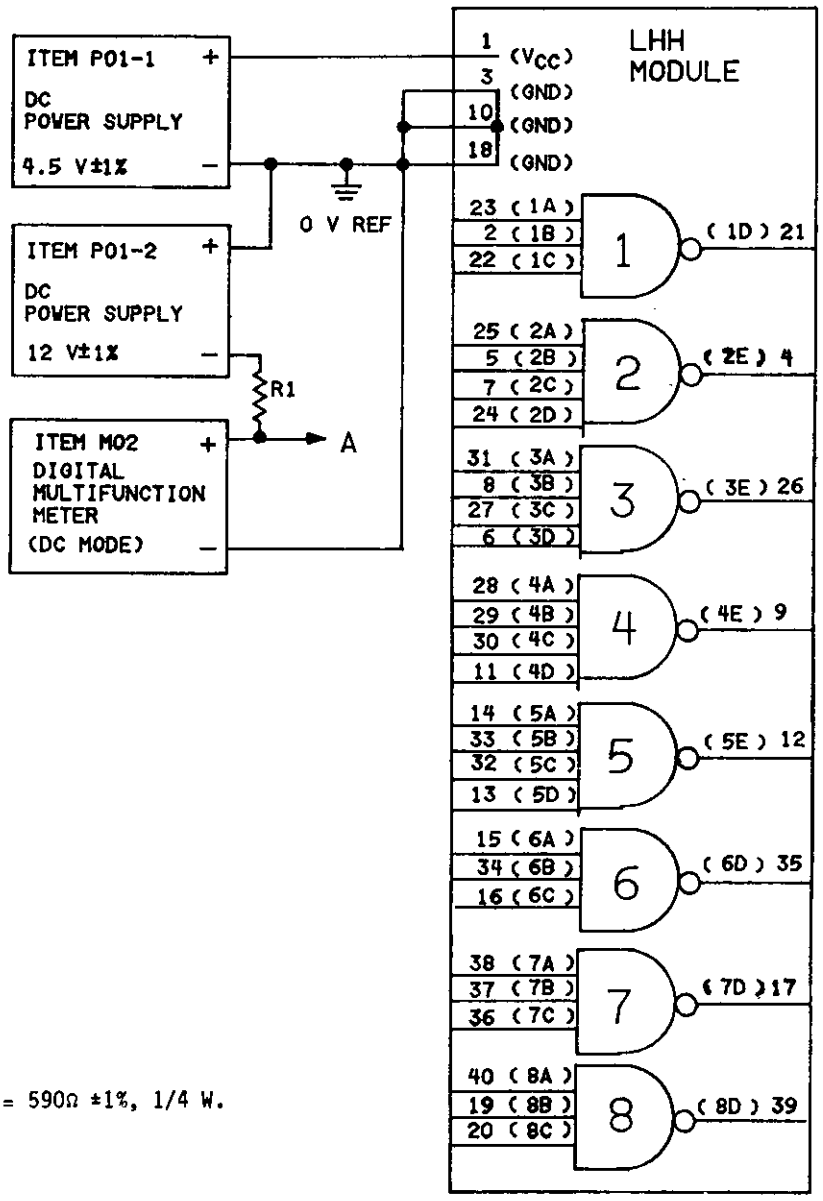
Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
F1	1A	Input clamp diode voltage	23	-0.4	-1.2	V	10
F2	1B	"	2	"	"	"	"
F3	1C	"	22	"	"	"	"
F4	2A	"	25	"	"	"	"
F5	2B	"	5	"	"	"	"
F6	2C	"	7	"	"	"	"
F7	2D	"	24	"	"	"	"
F8	3A	"	31	"	"	"	"
F9	3B	"	8	"	"	"	"
F10	3C	"	27	"	"	"	"
F11	3D	"	6	"	"	"	"
F12	4A	"	28	"	"	"	"
F13	4B	"	29	"	"	"	"
F14	4C	"	30	"	"	"	"
F15	4D	"	11	"	"	"	"
F16	5A	"	14	"	"	"	"
F17	5B	"	33	"	"	"	"
F18	5C	"	32	"	"	"	"
F19	5D	"	13	"	"	"	"
F20	6A	"	15	"	"	"	"
F21	6B	"	34	"	"	"	"

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*TABLE VIII. Input clamp diode voltage test - Continued.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
F22	6C Input clamp diode voltage	16	-0.4	-1.2	V	10
F23	7A "	38	"	"	"	"
F24	7B "	37	"	"	"	"
F25	7C "	36	"	"	"	"
F26	8A "	40	"	"	"	"
F27	8B "	19	"	"	"	"
F28	8C "	20	"	"	"	"

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NOTE: R1 = 590Ω ±1%, 1/4 W.

*FIGURE 10. Input clamp diode voltage test circuit for table VIII.

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*TABLE IX. Output voltage (low-level) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
G1	1D Output voltage, low level	2,22,23	Open	21	0	500	mV	11
G2	2E "	5,7,24,25	"	4	"	"	"	"
G3	3E "	6,8,27,31	"	26	"	"	"	"
G4	4E "	28,29,30,11	"	9	"	"	"	"
G5	5E "	13,14,32,33	"	12	"	"	"	"
G6	6D "	15,16,34	"	35	"	"	"	"
G7	7D "	36,37,38	"	17	"	"	"	"
G8	8D "	19,20,40	"	39	"	"	"	"

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*TABLE X. Output voltage (high-level) test.

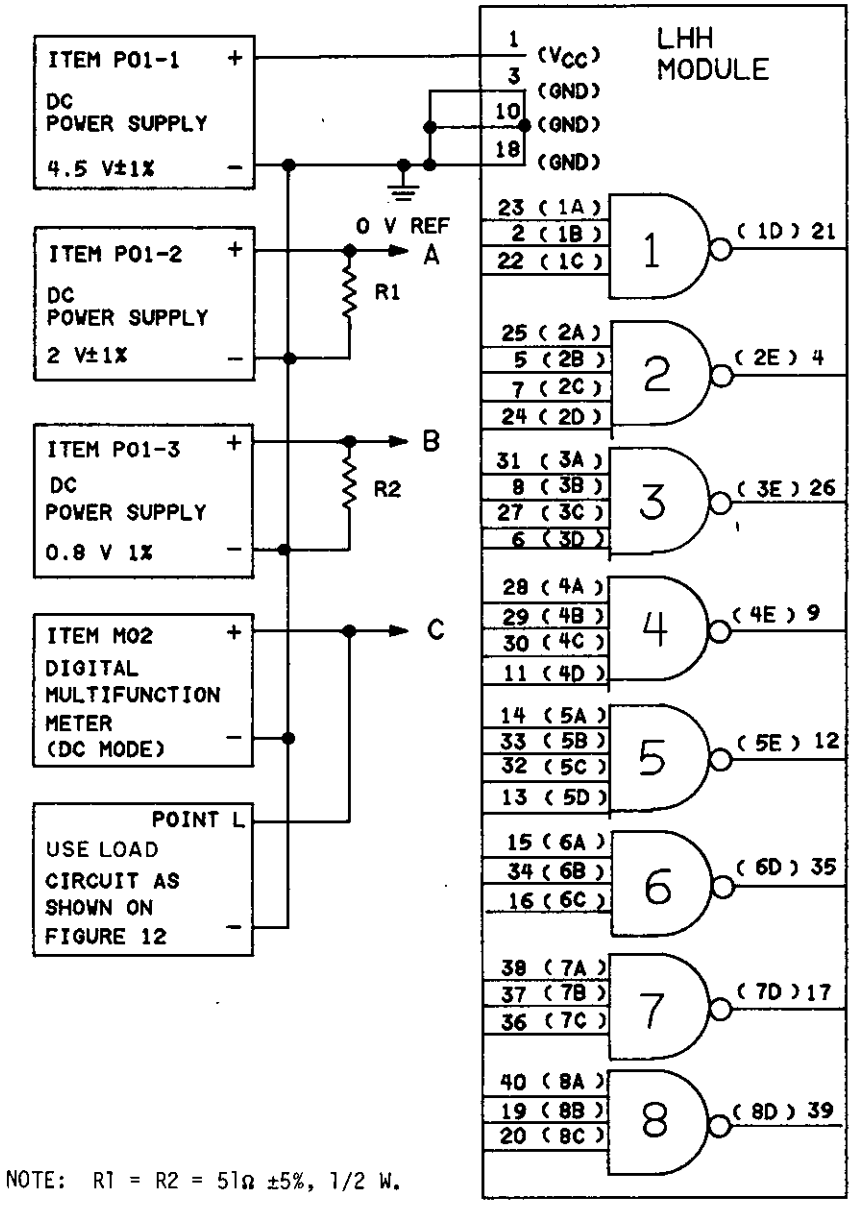
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H1	1D	Output voltage, high level	22,23	2	21	2.5	4.5	V	11
H2	1D	"	22,2	23	"	"	"	"	"
H3	1D	"	2,23	22	"	"	"	"	"
H4	2E	"	7,24,25	5	4	"	"	"	"
H5	2E	"	5,24,25	7	"	"	"	"	"
H6	2E	"	5,7,25	24	"	"	"	"	"
H7	2E	"	5,7,24	25	"	"	"	"	"
H8	3E	"	8,27,31	6	26	"	"	"	"
H9	3E	"	6,27,31	8	"	"	"	"	"
H10	3E	"	6,8,31	27	"	"	"	"	"
H11	3E	"	6,8,27	31	"	"	"	"	"
H12	4E	"	28,29,30	11	9	"	"	"	"
H13	4E	"	11,29,30	28	"	"	"	"	"
H14	4E	"	11,28,30	29	"	"	"	"	"
H15	4E	"	11,28,29	30	"	"	"	"	"
H16	5E	"	13,14,33	32	12	"	"	"	"
H17	5E	"	13,14,32	33	"	"	"	"	"
H18	5E	"	14,33,32	13	"	"	"	"	"
H19	5E	"	13,33,32	14	"	"	"	"	"
H20	6D	"	15,16	34	35	"	"	"	"
H21	6D	"	15,34	16	35	"	"	"	"

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*TABLE X. Output voltage (high-level) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
H22	6D	Output voltage, high level	16,34	15	35	2.5	4.5	V	11
H23	7D	"	37,38	36	17	"	"	"	"
H24	7D	"	36,38	37	"	"	"	"	"
H25	7D	"	36,37	38	"	"	"	"	"
H26	8D	"	20,40	19	39	"	"	"	"
H27	8D	"	19,40	20	"	"	"	"	"
H28	8D	"	19,20	40	"	"	"	"	"

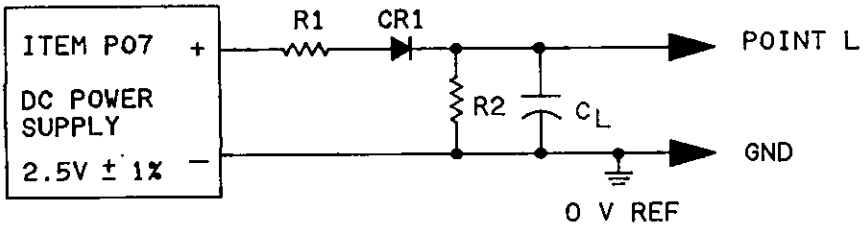
Source: https://assist.dla.mil -- Downloaded: 2026-10-20T15:51Z
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

*FIGURE 11. Output voltage (low-level and high-level) test circuit for tables IX and X.

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NOTES:
1. CR1 = 1N3600, 1N4150, or equivalent.
2. R1 = Select value for a current flow of 20 mA ±5% out of point L with point L held at 0.5 V.
3. R2 = 2.49 kΩ ±1%, 1/8 W.
4. Test type S:
   C_L = 50 ±5 pF (including probe and parasitic capacitance).
   Test type Q:
   C_L = 150 ±5 pF (including probe and parasitic capacitance) except for test conditions U1-U8 and V1-V8 which will use 50 ±5 pF.
5. Capacitor C_L is not required for test conditions G1-G8 and H1-H28.
6. Lead length from load circuit to module output pin under test is to be less than 1 inch (25.4 mm).

FIGURE 12. LHH logic load circuit.

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TABLE XI. Input current (high-level) test.

Test condition		Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J1	1A	Input current, high level	23	2,5,6,7,8,11,13,14,15,16, 19,20,22,24,25,27,28,29,30 31,32,33,34,36,37,38,40	0	100	µA	13
J2	1B	"	2	5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	50	"	"
J3	1C	"	22	2,5,6,7,8,11,13,14,15,16, 19,20,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J4	2A	"	25	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J5	2B	"	5	2,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J6	2C	"	7	2,5,6,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J7	2D	"	24	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J8	3A	"	31	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,32,33,34,36,37,38,40	"	"	"	"
J9	3B	"	8	2,5,6,7,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J10	3C	"	27	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J11	3D	"	6	2,5,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"

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TABLE XI. Input current (high-level) test - Continued.

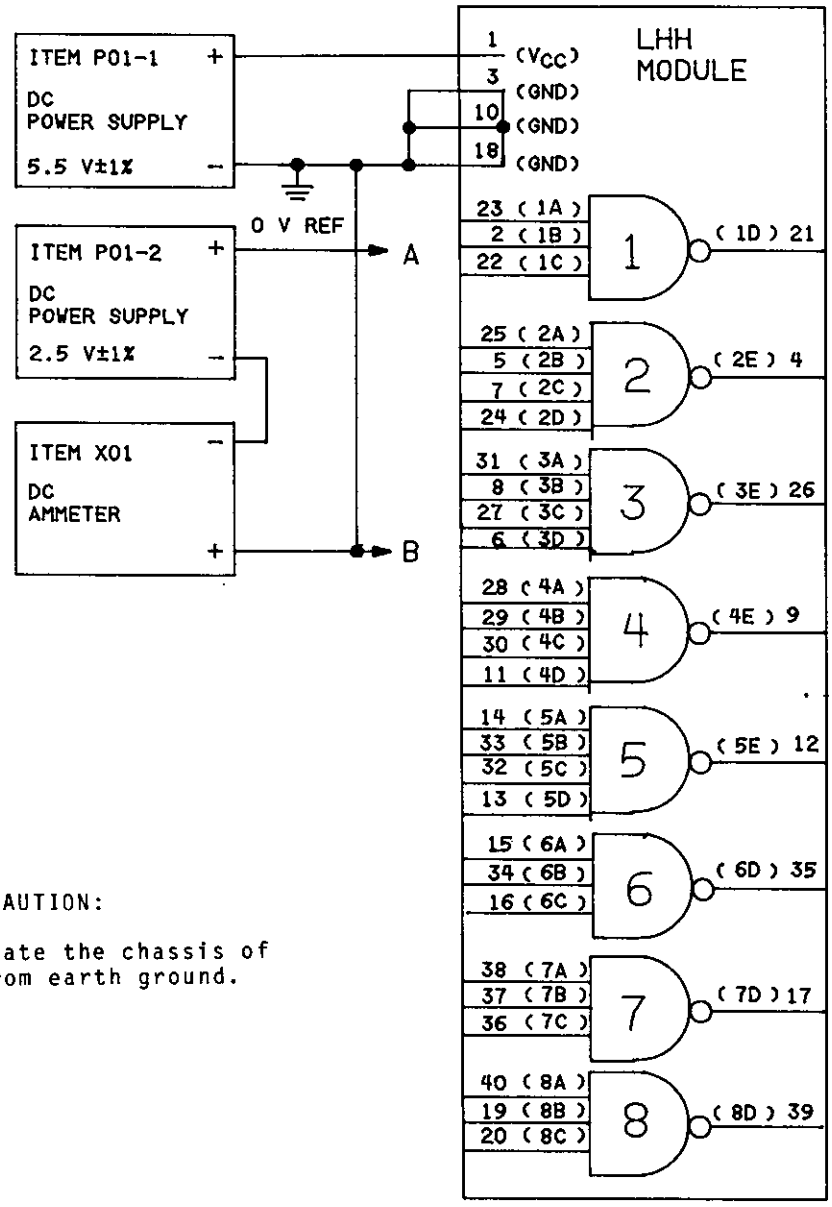
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
J12	4A Input curent, high level	28	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,29, 30,31,32,33,34,36,37,38,40	0	50	µA	13
J13	4B "	29	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28, 30,31,32,33,34,36,37,38,40	"	"	"	"
J14	4C "	30	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28, 29,31,32,33,34,36,37,38,40	"	"	"	"
J15	4D "	11	2,5,6,7,8,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J16	5A "	14	2,5,6,7,8,11,13,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J17	5B "	33	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,34,36,37,38,40	"	"	"	"
J18	5C "	32	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,33,34,36,37,38,40	"	"	"	"
J19	5D "	13	2,5,6,7,8,11,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J20	6A "	15	2,5,6,7,8,11,13,14,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J21	6B "	34	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,36,37,38,40	"	"	"	"
J22	6C "	16	2,5,6,7,8,11,13,14,15, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	100	"	"

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TABLE XI. Input current (high-level) test - Continued.

Test condition	Parameter		Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	Min	Max		
J23	7A	Input current, high level	38	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,40	0	100	µA	13
J24	7B	"	37	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,38,40	"	50	"	"
J25	7C	"	36	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,37,38,40	"	"	"	"
J26	8A	"	40	2,5,6,7,8,11,13,14,15,16, 19,20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38	"	"	"	"
J27	8B	"	19	2,5,6,7,8,11,13,14,15,16, 20,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	"	"	"
J28	8C	"	20	2,5,6,7,8,11,13,14,15,16, 19,22,23,24,25,27,28,29, 30,31,32,33,34,36,37,38,40	"	100	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 13. Input current (high-level) test circuit for table XI.

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TABLE XII. Input current (low-level) test.

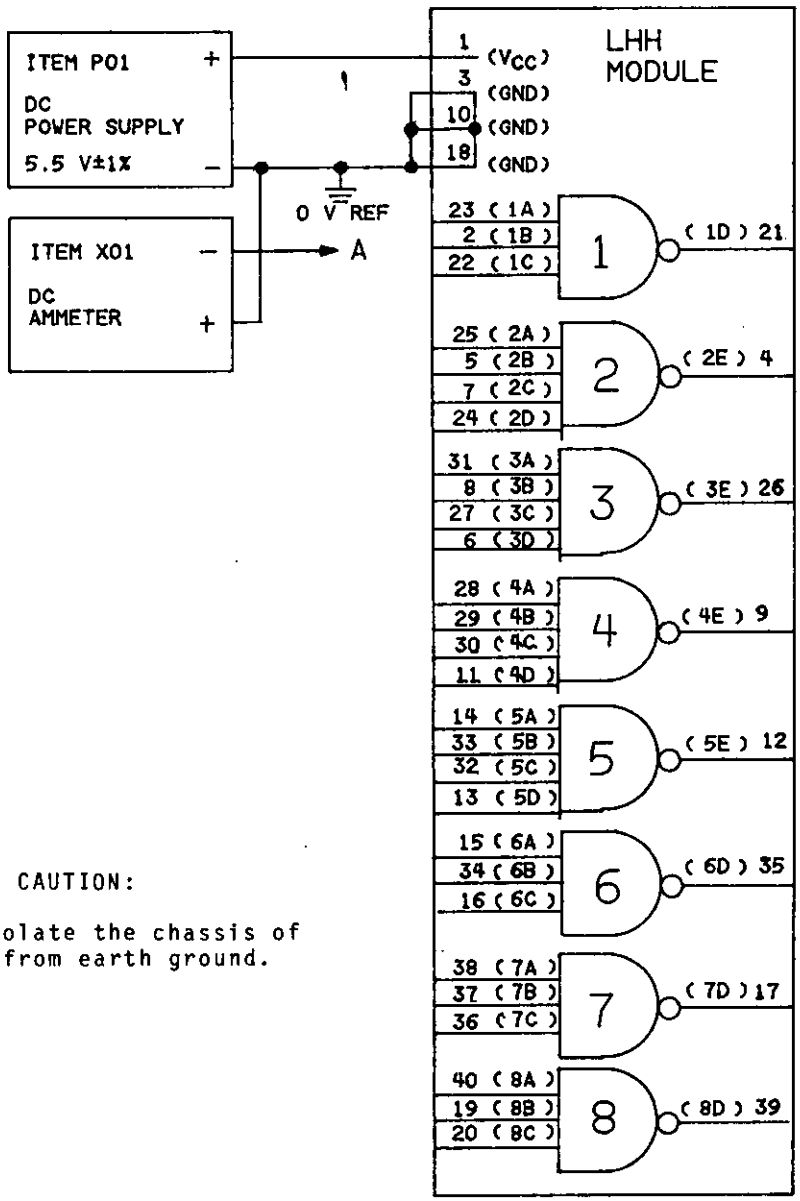
Test condition		Parameter	Connect A to pin	Initial limits 25°C ±5°C ambient Min	Initial limits 25°C ±5°C ambient Max	Unit	Test figure
K1	1A	Input current, low level	23	-0.4	-2.2	mA	14
K2	1B	"	2	"	"	"	"
K3	1C	"	22	"	"	"	"
K4	2A	"	25	"	"	"	"
K5	2B	"	5	"	"	"	"
K6	2C	"	7	"	"	"	"
K7	2D	"	24	"	"	"	"
K8	3A	"	31	"	"	"	"
K9	3B	"	8	"	"	"	"
K10	3C	"	27	"	"	"	"
K11	3D	"	6	"	"	"	"
K12	4A	"	28	"	"	"	"
K13	4B	"	29	"	"	"	"
K14	4C	"	30	"	"	"	"
K15	4D	"	11	"	"	"	"
K16	5A	"	14	"	"	"	"
K17	5B	"	33	"	"	"	"
K18	5C	"	32	"	"	"	"
K19	5D	"	13	"	"	"	"
K20	6A	"	15	"	"	"	"
K21	6B	"	34	"	"	"	"

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TABLE XII. Input current (low-level) test - Continued.

Test condition	Parameter		Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	Min	Max		
K22	6C	Input current, low level	16	-0.4	-2.2	mA	14
K23	7A	"	38	"	"	"	"
K24	7B	"	37	"	"	"	"
K25	7C	"	36	"	"	"	"
K26	8A	"	40	"	"	"	"
K27	8B	"	19	"	"	"	"
K28	8C	"	20	"	"	"	"

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CAUTION:

Do not isolate the chassis of item X01 from earth ground.

FIGURE 14. Input current (low-level) test circuit for table XII.

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*TABLE XIII. Continuity test.

Test condition	Parameter	Connect	Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	Min	Max		
L1	Continuity	3	0	1	Ω	15
L2	"	18	0	1	"	"
L3	"	3	0.0	0.1	"	"
L4	"	18	0.0	0.1	"	"

TABLE XIV. Power supply current (low-level and high-level outputs) test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to point	B to point	Min	Max		
M1	Power supply current, low-level outputs	C	Open	30	72	mA	16
M2	Power supply current, high-level outputs	Open	C	15	32	mA	16

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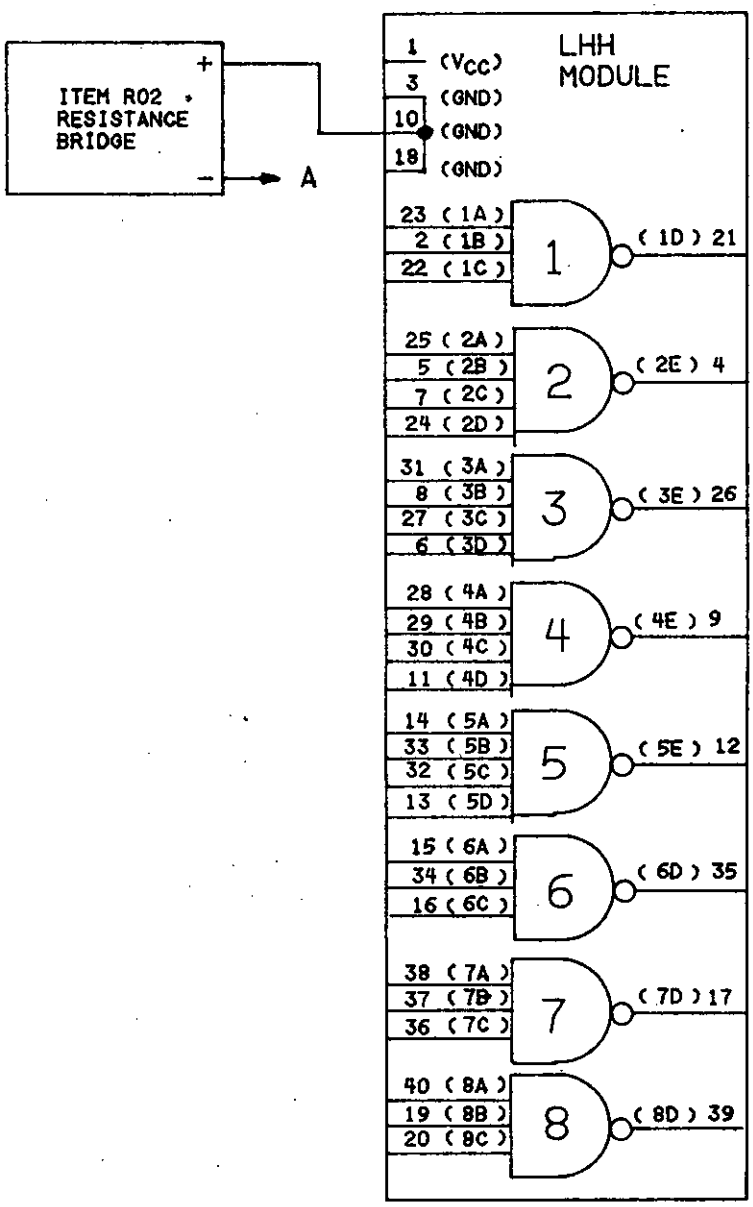


FIGURE 15. Continuity test circuit for table XIII.

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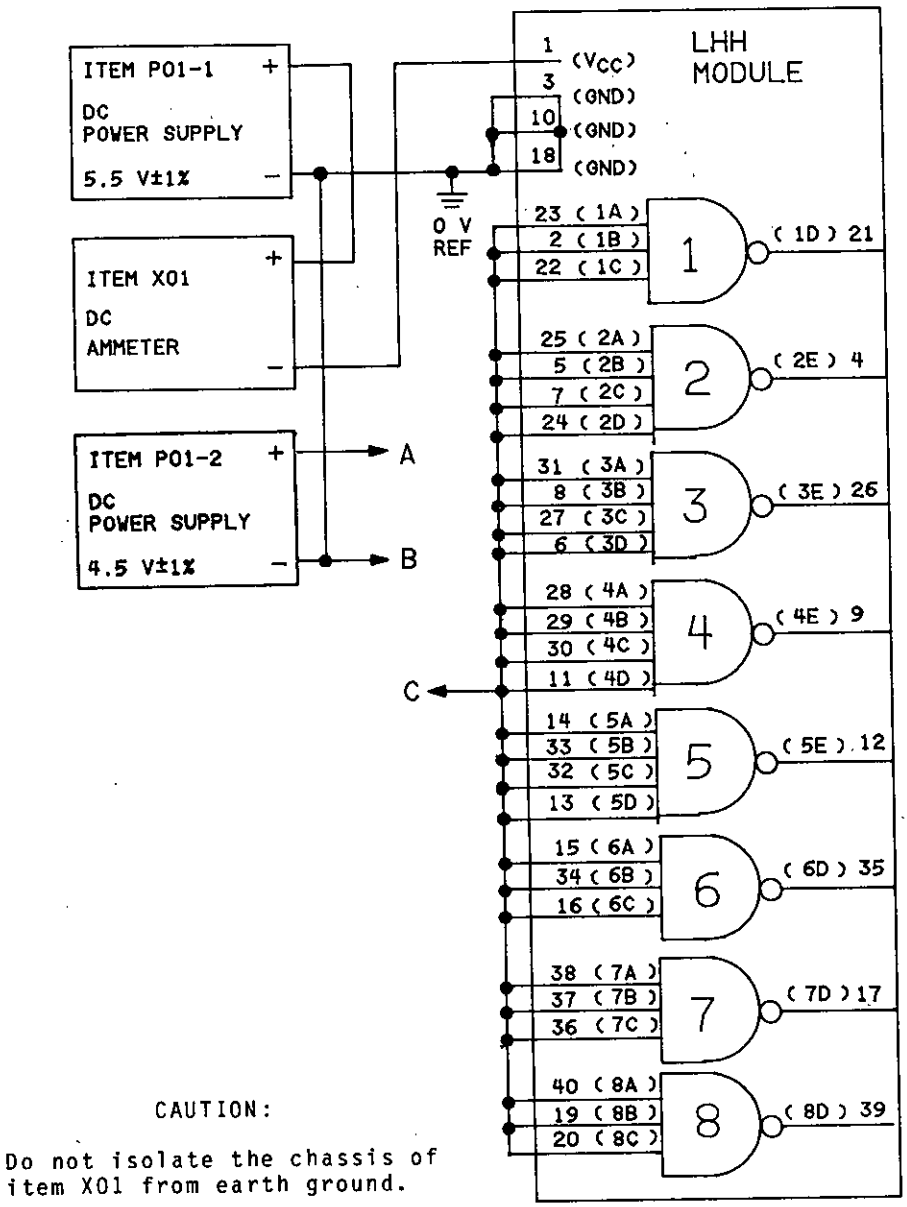


FIGURE 16. Power supply current (low-level and high-level outputs) test circuit for table XIV.

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test.

Test condition	Parameter	Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	Min	Max		
N1	1D Transition time (t_{TLH}), C_L = 50 pF	22,23	2	21	2	15	ns	17,18
N2	2E "	24,25,7	5	4	"	"	"	"
N3	3E "	27,31,8	6	26	"	"	"	"
N4	4E "	28,29,30	11	9	"	"	"	"
N5	5E "	32,33,14	13	12	"	"	"	"
N6	6D "	34,16	15	35	"	"	"	"
N7	7D "	37,38	36	17	"	"	"	"
N8	8D "	20,40	19	39	"	"	"	"
N9	1D Transition time (t_{TLH}), C_L = 150 pF	22,23	2	21	5	27	"	"
N10	2E "	24,25,7	5	4	"	"	"	"
N11	3E "	27,31,8	6	26	"	"	"	"
N12	4E "	28,29,30	11	9	"	"	"	"
N13	5E "	32,33,14	13	12	"	"	"	"
N14	6D "	34,16	15	35	"	"	"	"
N15	7D "	37,38	36	17	"	"	"	"
N16	8D "	20,40	19	39	"	"	"	"
P1	1D Transition time (t_{THL}), C_L = 50 pF	22,23	2	21	2	6	"	"
P2	2E "	24,25,7	5	4	"	"	"	"
P3	3E "	27,31,8	6	26	"	"	"	"
P4	4E "	28,29,30	11	9	"	"	"	"
P5	5E "	32,33,14	13	12	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
P6	6D	Transition time (t_{THL}), C_L = 50 pF	34,16	15	35	2	6	ns	17,18
P7	7D	"	37,38	36	17	"	"	"	"
P8	8D	"	20,40	19	39	"	"	"	"
P9	1D	Transition time (t_{THL}), C_L = 150 pF	22,23	2	21	3	11	"	"
P10	2E	"	24,25,7	5	4	"	"	"	"
P11	3E	"	27,31,8	6	26	"	"	"	"
P12	4E	"	28,29,30	11	9	"	"	"	"
P13	5E	"	32,33,14	13	12	"	"	"	"
P14	6D	"	34,16	15	35	"	"	"	"
P15	7D	"	37,38	36	17	"	"	"	"
P16	8D	"	20,40	19	39	"	"	"	"
Q1	1A to 1D	Propagation delay time (t_{PLH}), C_L = 50 pF	2,22	23	21	2	7	"	"
Q2	1B to 1D	"	22,23	2	"	"	"	"	"
Q3	1C to 1D	"	2,23	22	"	"	"	"	"
Q4	2A to 2E	"	5,7,24	25	4	"	"	"	"
Q5	2B to 2E	"	7,24,25	5	"	"	"	"	"
Q6	2C to 2E	"	5,24,25	7	"	"	"	"	"
Q7	2D to 2E	"	5,7,25	24	"	"	"	"	"
Q8	3A to 3E	"	8,27,6	31	26	"	"	"	"
Q9	3B to 3E	"	6,27,31	8	"	"	"	"	"
Q10	3C to 3E	"	6,8,31	27	"	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q11	3D to 3E	Propagation delay time (t_{PLH}), C_L = 50 pF	8,27,31	6	26	2	7	ns	17,18
Q12	4A to 4E	"	29,30,11	28	9	"	"	"	"
Q13	4B to 4E	"	28,30,11	29	"	"	"	"	"
Q14	4C to 4E	"	28,29,11	30	"	"	"	"	"
Q15	4D to 4E	"	28,29,30	11	"	"	"	"	"
Q16	5A to 5E	"	13,33,32	14	12	"	"	"	"
Q17	5B to 5E	"	13,14,32	33	"	"	"	"	"
Q18	5C to 5E	"	13,14,33	32	"	"	"	"	"
Q19	5D to 5E	"	14,33,32	13	"	"	"	"	"
Q20	6A to 6D	"	16,34	15	35	"	"	"	"
Q21	6B to 6D	"	16,15	34	"	"	"	"	"
Q22	6C to 6D	"	15,34	16	"	"	"	"	"
Q23	7A to 7D	"	37,36	38	17	"	"	"	"
Q24	7B to 7D	"	38,36	37	"	"	"	"	"
Q25	7C to 7D	"	37,38	36	"	"	"	"	"
Q26	8A to 8D	"	19,20	40	39	"	"	"	"
Q27	8B to 8D	"	20,40	19	"	"	"	"	"
Q28	8C to 8D	"	19,40	20	"	"	"	"	"
Q29	1A to 1D	Propagation delay time (t_{PLH}), C_L = 150 pF	2,22	23	21	3.0	10.5	"	"
Q30	1B to 1D	"	22,23	2	"	"	"	"	"
Q31	1C to 1D	"	2,23	22	"	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q32	2A to 2E	Propagation delay time (t_{PLH}), C_L = 150 pF	5,7,24	25	4	3.0	10.5	ns	17,18
Q33	2B to 2E	"	7,24,25	5	"	"	"	"	"
Q34	2C to 2E	"	5,24,25	7	"	"	"	"	"
Q35	2D to 2E	"	5,7,25	24	"	"	"	"	"
Q36	3A to 3E	"	8,27,6	31	26	"	"	"	"
Q37	3B to 3E	"	31,27,6	8	"	"	"	"	"
Q38	3C to 3E	"	31,8,6	27	"	"	"	"	"
Q39	3D to 3E	"	31,8,27	6	"	"	"	"	"
Q40	4A to 4E	"	29,30,11	28	9	"	"	"	"
Q41	4B to 4E	"	28,30,11	29	"	"	"	"	"
Q42	4C to 4E	"	28,29,11	30	"	"	"	"	"
Q43	4D to 4E	"	28,29,30	11	"	"	"	"	"
Q44	5A to 5E	"	13,33,32	14	12	"	"	"	"
Q45	5B to 5E	"	13,14,32	33	"	"	"	"	"
Q46	5C to 5E	"	13,14,33	32	"	"	"	"	"
Q47	5D to 5E	"	14,33,32	13	"	"	"	"	"
Q48	6A to 6D	"	34,16	15	35	"	"	"	"
Q49	6B to 6D	"	15,16	34	"	"	"	"	"
Q50	6C to 6D	"	15,34	16	"	"	"	"	"
Q51	7A to 7D	"	36,37	38	17	"	"	"	"
Q52	7B to 7D	"	36,38	37	"	"	"	"	"
Q53	7C to 7D	"	37,38	36	"	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
Q54	8A to 8D	Propagation delay time (t_{PLH}), C_L = 150 pF	19,20	40	39	3.0	10.5	ns	17,18
Q55	8B to 8D	"	20,40	19	"	"	"	"	"
Q56	8C to 8D	"	19,40	20	"	"	"	"	"
R1	1A to 1D	Propagation delay time (t_{PHL}), C_L = 50 pF	2,22	23	21	2.0	7.5	"	"
R2	1B to 1D	"	22,23	2	"	"	"	"	"
R3	1C to 1D	"	2,23	22	"	"	"	"	"
R4	2A to 2E	"	5,7,24	25	4	"	"	"	"
R5	2B to 2E	"	7,24,25	5	"	"	"	"	"
R6	2C to 2E	"	5,24,25	7	"	"	"	"	"
R7	2D to 2E	"	5,7,25	24	"	"	"	"	"
R8	3A to 3E	"	8,27,6	31	26	"	"	"	"
R9	3B to 3E	"	6,27,31	8	"	"	"	"	"
R10	3C to 3E	"	6,8,31	27	"	"	"	"	"
R11	3D to 3E	"	8,27,31	6	"	"	"	"	"
R12	4A to 4E	"	29,30,11	28	9	"	"	"	"
R13	4B to 4E	"	28,30,11	29	"	"	"	"	"
R14	4C to 4E	"	28,29,11	30	"	"	"	"	"
R15	4D to 4E	"	28,29,30	11	"	"	"	"	"
R16	5A to 5E	"	13,33,32	14	12	"	"	"	"
R17	5B to 5E	"	13,14,32	33	"	"	"	"	"
R18	5C to 5E	"	13,14,33	32	"	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

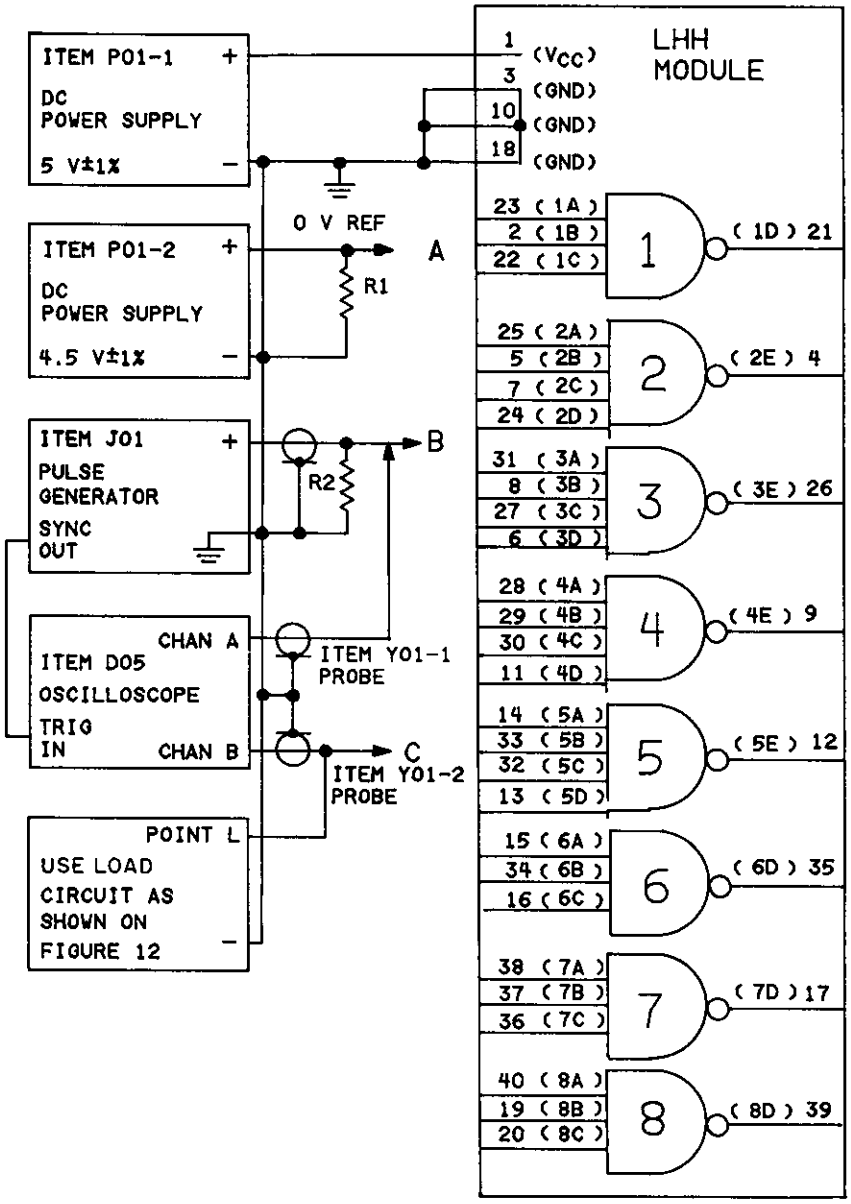
Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R19	5D to 5E	Propagation delay time (t_{PHL}), C_L = 50 pF	14,33,32	13	12	2.0	7.5	ns	17,18
R20	6A to 6D	"	16,34	15	35	"	"	"	"
R21	6B to 6D	"	15,16	34	"	"	"	"	"
R22	6C to 6D	"	15,34	16	"	"	"	"	"
R23	7A to 7D	"	37,36	38	17	"	"	"	"
R24	7B to 7D	"	38,36	37	"	"	"	"	"
R25	7C to 7D	"	37,38	36	"	"	"	"	"
R26	8A to 8D	"	19,20	40	39	"	"	"	"
R27	8B to 8D	"	20,40	19	"	"	"	"	"
R28	8C to 8D	"	19,40	20	"	"	"	"	"
R29	1A to 1D	Propagation delay time (t_{PHL}), C_L = 150 pF	2,22	23	21	3.0	11.5	"	"
R30	1B to 1D	"	22,23	2	"	"	"	"	"
R31	1C to 1D	"	2,23	22	"	"	"	"	"
R32	2A to 2E	"	5,7,24	25	4	"	"	"	"
R33	2B to 2E	"	7,24,25	5	"	"	"	"	"
R34	2C to 2E	"	5,24,25	7	"	"	"	"	"
R35	2D to 2E	"	5,7,25	24	"	"	"	"	"
R36	3A to 3E	"	8,27,6	31	26	"	"	"	"
R37	3B to 3E	"	31,27,6	8	"	"	"	"	"
R38	3C to 3E	"	31,8,6	27	"	"	"	"	"
R39	3D to 3E	"	31,8,27	6	"	"	"	"	"

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*TABLE XV. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test - Continued.

Test condition	Parameter		Connect			Initial limits 25°C ±5°C ambient		Unit	Test figure
			A to pin	B to pin	C to pin	Min	Max		
R40	4A to 4E	Propagation delay time (t_{PHL}), C_L = 150 pF	29,30,11	28	9	3.0	11.5	ns	17,18
R41	4B to 4E	"	28,30,11	29	"	"	"	"	"
R42	4C to 4E	"	28,29,11	30	"	"	"	"	"
R43	4D to 4E	"	28,29,30	11	"	"	"	"	"
R44	5A to 5E	"	13,33,32	14	12	"	"	"	"
R45	5B to 5E	"	13,14,32	33	"	"	"	"	"
R46	5C to 5E	"	13,14,33	32	"	"	"	"	"
R47	5D to 5E	"	14,33,32	13	"	"	"	"	"
R48	6A to 6D	"	34,16	15	35	"	"	"	"
R49	6B to 6D	"	15,16	34	"	"	"	"	"
R50	6C to 6D	"	15,34	16	"	"	"	"	"
R51	7A to 7D	"	36,37	38	17	"	"	"	"
R52	7B to 7D	"	36,38	37	"	"	"	"	"
R53	7C to 7D	"	37,38	36	"	"	"	"	"
R54	8A to 8D	"	19,20	40	39	"	"	"	"
R55	8B to 8D	"	20,40	19	"	"	"	"	"
R56	8C to 8D	"	19,40	20	"	"	"	"	"

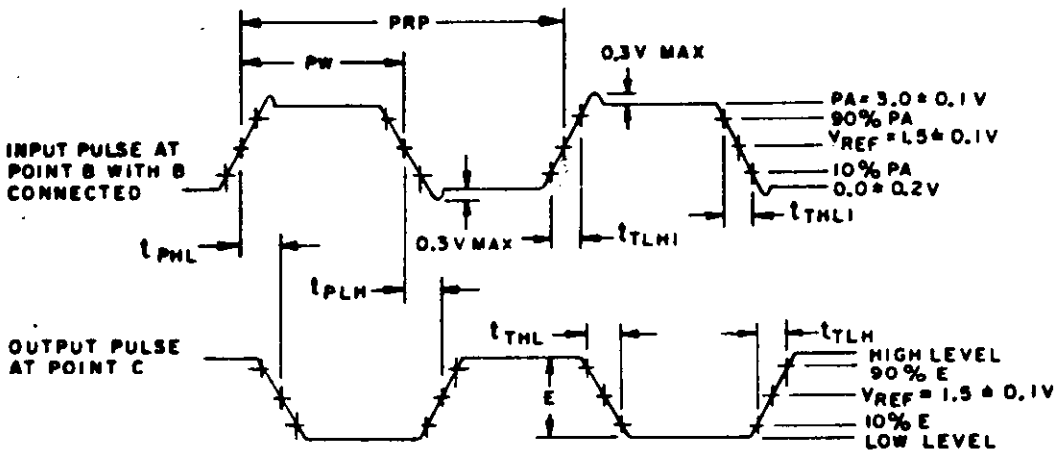
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NOTE: R1 = R2 = 51Ω ±5%, 1/2 W.

*FIGURE 17. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit for table XV.

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NOTES:
1. Input pulse parameters:
   $t_{TLH1} = t_{THL1} = 4 \pm 1$ ns.
   PW = 100 ±10 ns.
   PRP = 500 ±50 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$2.5 V and the low level is $\leq$0.5 V.
3. The lead length of electrical conductors between resistor R2 and input pins; between items Y01-1 and Y01-2 and input and output pins; and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

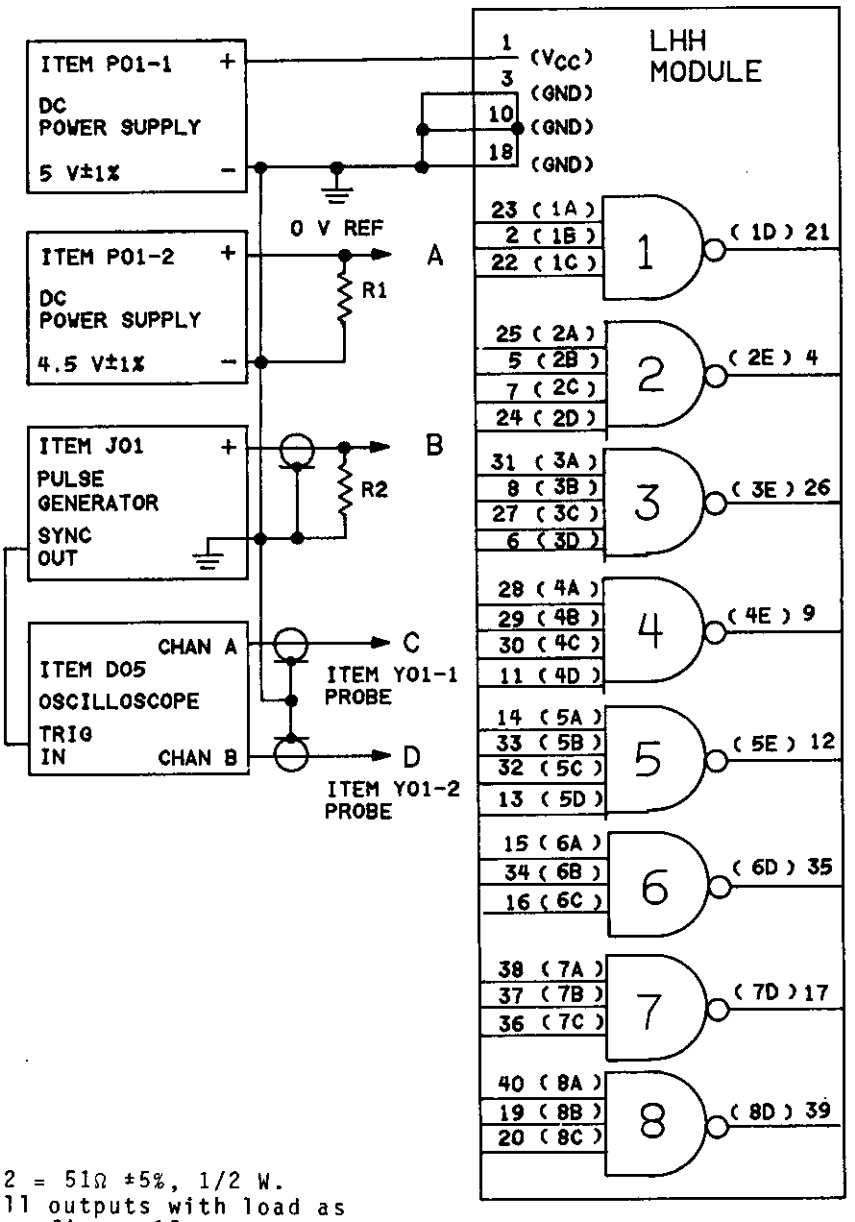
*FIGURE 18. Transition time (t_{TLH} and t_{THL}) and propagation delay time (t_{PLH} and t_{PHL}) test circuit waveforms for table XV.

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TABLE XVI. Output to output delta propagation delay time (Δt_{PLH} and Δt_{PHL}) test.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
S1	1D to 2E Output to output delta propagation delay time (Δt_{PLH}), C_L = 50 pF	23,2,24,5,7	22,25	21	4	-3	3	ns	19,20
S2	3E to 4E "	6,8,27,28, 29,11	30,11	9	26	"	"	"	"
S3	5E to 6D "	32,33,13, 34,16	14,15	12	35	"	"	"	"
S4	7D to 8D "	37,36,19, 20	38,40	17	39	"	"	"	"
T1	1D to 2E Output to output delta propagation delay time (Δt_{PHL}), C_L = 50 pF	23,2,24,5,7	22,25	21	4	"	"	"	"
T2	3E to 4E "	6,8,27,28, 29,11	30,31	9	26	"	"	"	"
T3	5E to 6D "	32,33,34, 13,16	14,15	12	35	"	"	"	"
T4	7D to 8D "	37,36,19, 20	38,40	17	39	"	"	"	"

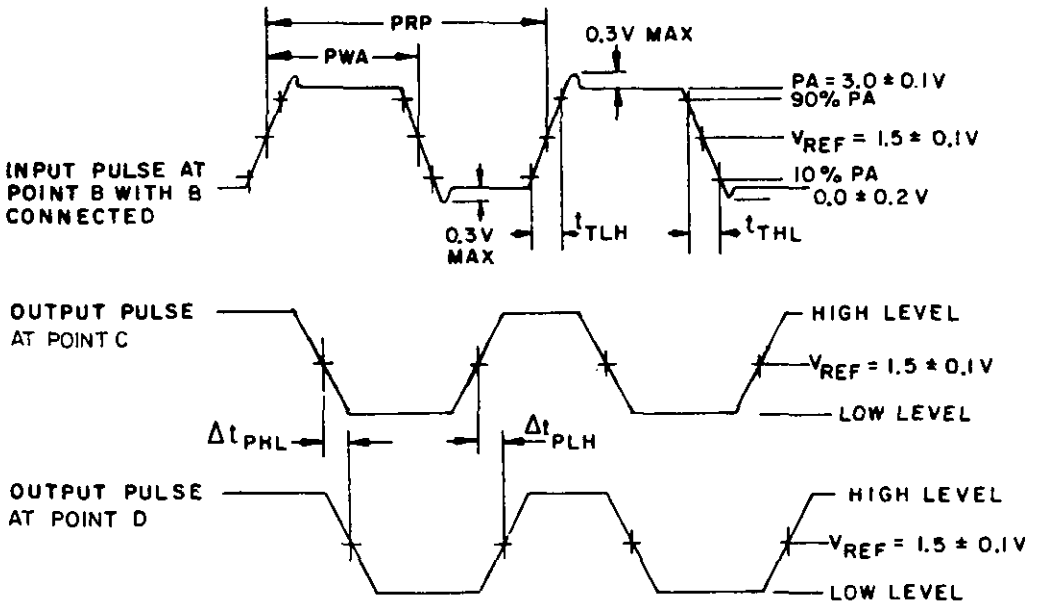
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NOTES:
1. R1 = R2 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as shown on figure 12.

FIGURE 19. Output to output delta propagation delay time (Δt_{PLH} and Δt_{PHL}) test circuit for table XVI.

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NOTES:
1. Input pulse parameters:
   $t_{TLH} = t_{THL} = 4 \pm 1$ ns.
   PWA = 100 ±10 ns.
   PRP = 500 ±50 ns.
2. Before making the measurements, verify that the output pulse high level is $\geq$2.5 V and the low level is $\leq$0.5 V.
3. The lead length of electrical conductors between resistor R2 and input pins; between items YO1-1 and YO1-2 and input and output pins; and between load circuits and output pins shall be less than 1 inch (25.4 mm).
4. All voltages are referenced to ground.

*FIGURE 20. Output to output delta propagation delay time (Δt_{PLH} and Δt_{PHL}) test circuit waveforms for table XVI.

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TABLE XVII. Output crosstalk (low-level and high-level) test.

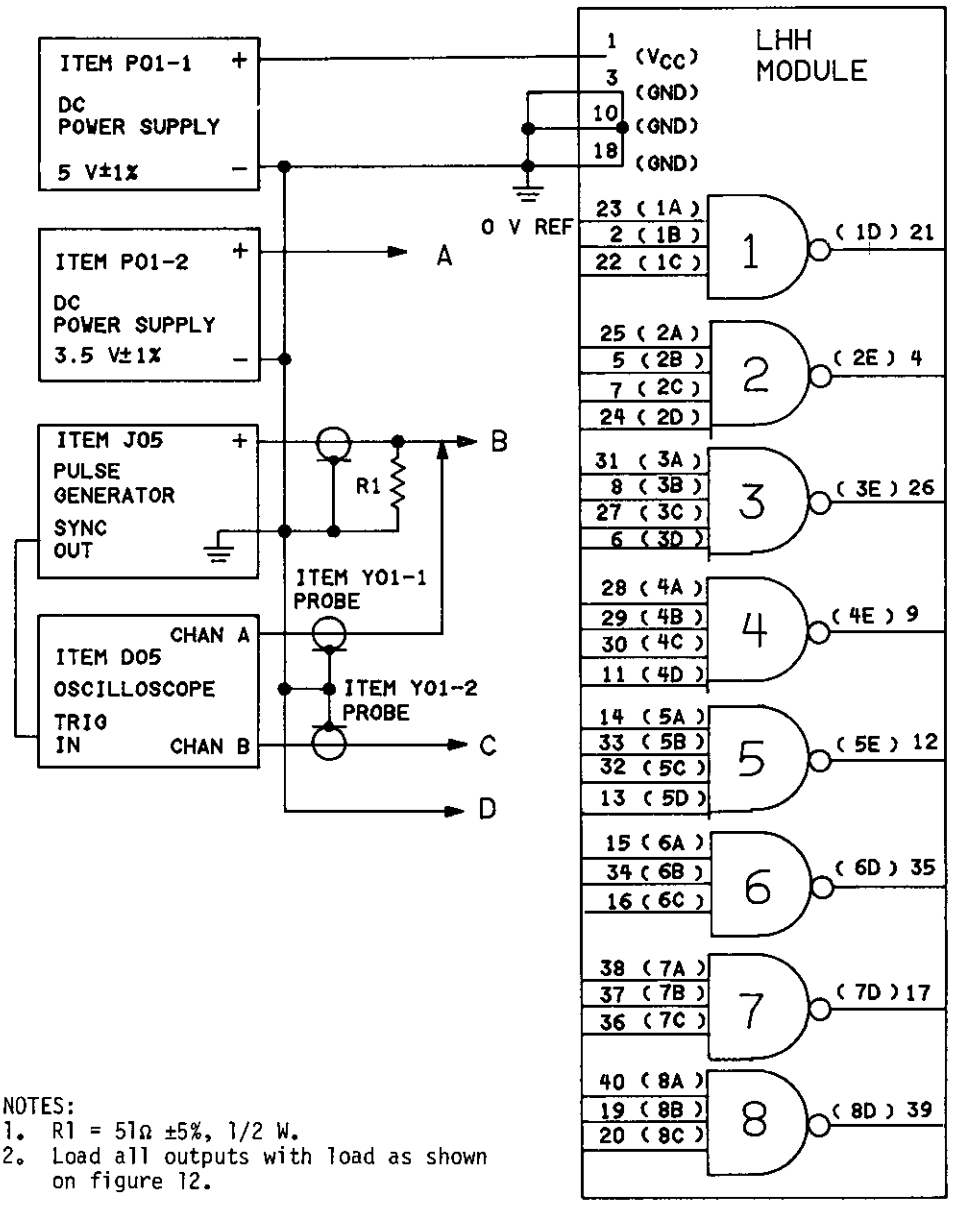
Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
U1	1D Output crosstalk, low level	2,22,23,5,7,24,8,27,6, 29,30,11,33,32,13,34, 16,37,36,19,20	25,31,28, 14,15,38, 40	21	Open	0	500	mV	21,22
U2	2E "	5,7,24,25,2,22,8,27,6, 29,30,11,33,32,13,34, 16,37,36,19,20	23,31,28, 14,15,38, 40	4	"	"	"	"	"
U3	3E "	6,8,27,31,2,22,5,7,24, 29,30,11,33,32,13,34, 16,37,36,19,20	23,25,28, 14,15,38, 40	26	"	"	"	"	"
U4	4E "	28,29,30,11,2,22,5,7, 24,8,27,6,33,32,13,34, 16,37,36,19,20	23,25,31, 14,15,38, 40	9	"	"	"	"	"
U5	5E "	13,14,32,33,2,22,5,7, 24,8,27,6,29,30,11,34, 16,37,36,19,20	23,25,31, 28,15,38, 40	12	"	"	"	"	"
U6	6D "	16,15,34,2,22,5,7,24, 8,27,6,29,30,11,33, 32,13,37,36,19,20	23,25,31, 28,14,38, 40	35	"	"	"	"	"
U7	7D "	36,37,38,2,22,5,7,24, 8,27,6,29,30,11,33, 32,13,34,16,19,20	23,25,31, 28,14,15, 40	17	"	"	"	"	"
U8	8D "	20,40,19,2,22,5,7,24, 8,27,6,29,30,11,33, 32,13,34,37,36,16	23,25,31, 28,14,15, 38	39	"	"	"	"	"
V1	1D Output crosstalk, high level	5,7,24,8,27,6,29,30, 11,33,32,13,34,16,37, 36,19,20	25,31,28, 14,15,38, 40	21	2,22,23	"	400	"	"
V2	2E "	2,22,8,27,6,29,30,11, 33,32,13,34,16,37,36, 19,20	23,31,28, 14,15,38, 40	4	24,25, 5,7	"	"	"	"
V3	3E "	2,22,5,7,24,29,30,11, 33,32,13,34,16,37,36, 19,20	23,25,28, 14,15,38, 40	26	6,8, 27,31	"	"	"	"

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TABLE XVII. Output crosstalk (low-level and high-level) test - Continued.

Test condition	Parameter	Connect				Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	Min	Max		
V4	4E Output crosstalk, high level	2,22,5,7,24,8,27,6, 33,32,13,34,16,37,36, 19,20	23,25,31, 14,15,38, 40	9	11,28, 29,30	0	400	mV	21,22
V5	5E "	2,22,5,7,24,8,27,6, 29,30,11,34,16,37,36, 19,20	23,25,31, 28,15,38, 40	12	13,14, 32,33	"	"	"	"
V6	6D "	2,22,5,7,24,8,27,6, 29,30,11,33,32,13,37, 36,19,20	23,25,31, 28,14,38, 40	35	15,16,34	"	"	"	"
V7	7D "	2,22,5,7,24,8,27,6, 29,30,11,33,32,13,34, 16,19,20	23,25,31, 28,14,15, 40	17	36,37,38	"	"	"	"
V8	8D "	2,22,5,7,24,8,27,6, 29,30,11,33,32,13,34, 16,37,36	23,25,31, 28,14,15, 38	39	19,20,40	"	"	"	"

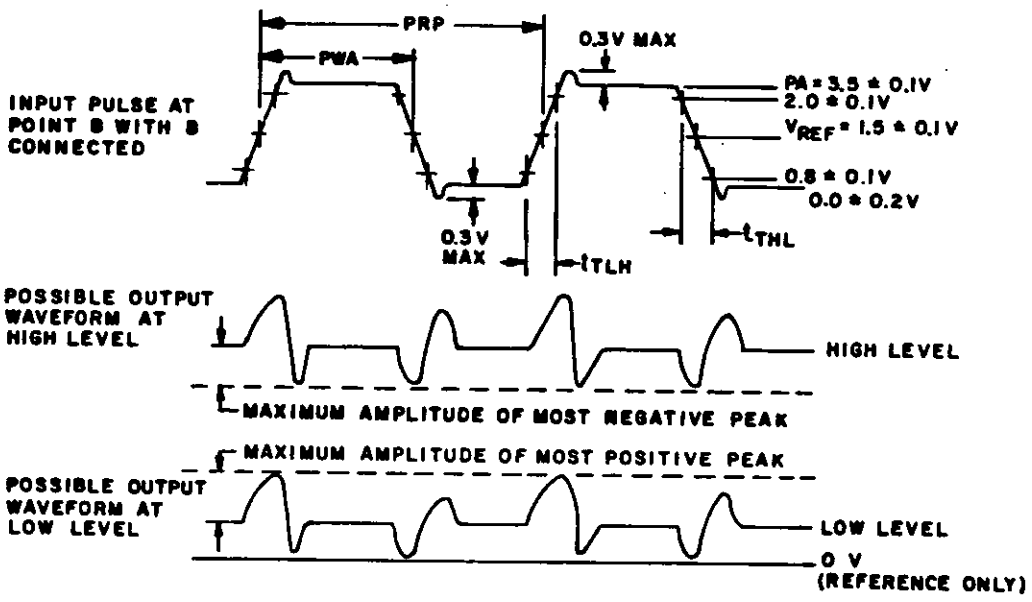
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NOTES:
1. R1 = 51Ω ±5%, 1/2 W.
2. Load all outputs with load as shown on figure 12.

FIGURE 21. Output crosstalk (low-level and high-level) test ciruit for table XVII.

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NOTES:
1. Input pulse parameters:
   $t_{THL} = t_{TLH} = 3.5 \pm 1.5$ ns (verified at each input pin).
   PWA = 250 ±25 ns.
   PRP = 500 ±50 ns.
2. The lead length of electrical conductors between resistor R1 and input pins; between items Y01-1 and Y01-2 and input and output load circuits and output pins; and between ground leads of items Y01-1 and Y01-2 and test fixture and module ground pins shall be less than 1 inch (25.4 mm).

FIGURE 22. Output crosstalk (low-level and high-level) test circuit waveforms for table XVII.

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TABLE XVIII. Power supply transient amplitude test.

Test condition	Parameter	Initial limits 25°C ±5°C ambient		Unit	Test figure
		Min	Max		
W	Power supply transient amplitude	0	400	mV	23,24

TABLE XIX. Power supply filter capacitance test.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
X	Power supply filter capacitance	Open	1	1.6	3.0	µF	25

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TABLE XX. Input capacitance test.

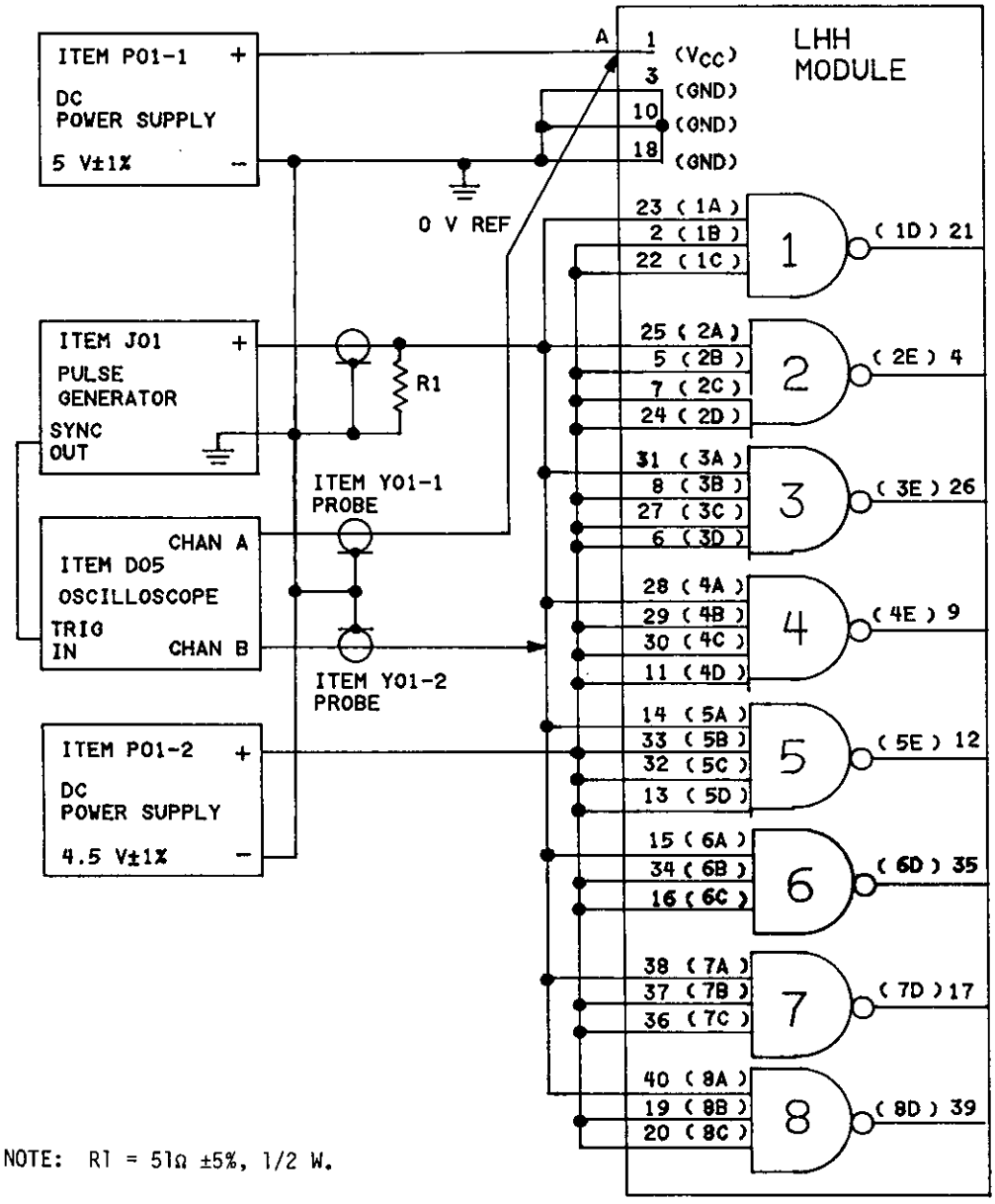
Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y1	1A Input capacitance	23	Open	5	30	pF	25
Y2	1B "	2	"	"	15	"	"
Y3	1C "	22	"	"	"	"	"
Y4	2A "	25	"	"	"	"	"
Y5	2B "	5	"	"	"	"	"
Y6	2C "	7	"	"	"	"	"
Y7	2D "	24	"	"	"	"	"
Y8	3A "	31	"	"	"	"	"
Y9	3B "	8	"	"	"	"	"
Y10	3C "	27	"	"	"	"	"
Y11	3D "	6	"	"	"	"	"
Y12	4A "	28	"	"	"	"	"
Y13	4B "	29	"	"	"	"	"
Y14	4C "	30	"	"	"	"	"
Y15	4D "	11	"	"	"	"	"
Y16	5A "	14	"	"	"	"	"
Y17	5B "	33	"	"	"	"	"
Y18	5C "	32	"	"	"	"	"
Y19	5D "	13	"	"	"	"	"
Y20	6A "	15	"	"	"	"	"
Y21	6B "	34	"	"	"	"	"

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TABLE XX. Input capacitance test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
Y22	6C Input capacitance	16	Open	5	30	pF	25
Y23	7A "	38	"	"	30	"	"
Y24	7B "	37	"	"	15	"	"
Y25	7C "	36	"	"	"	"	"
Y26	8A "	40	"	"	"	"	"
Y27	8B "	19	"	"	"	"	"
Y28	8C "	20	"	"	30	"	"

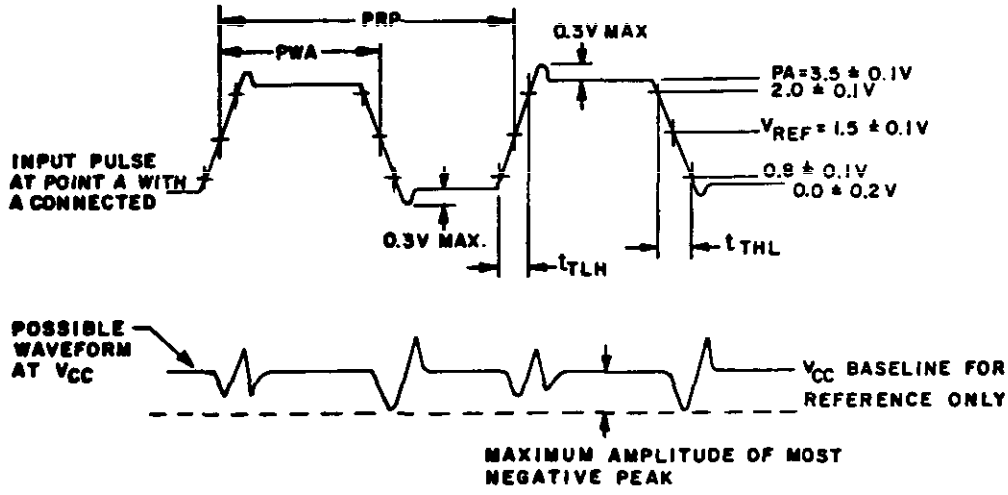
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NOTE: R1 = 51Ω ±5%, 1/2 W.

FIGURE 23. Power supply transient amplitude test circuit for table XVIII.

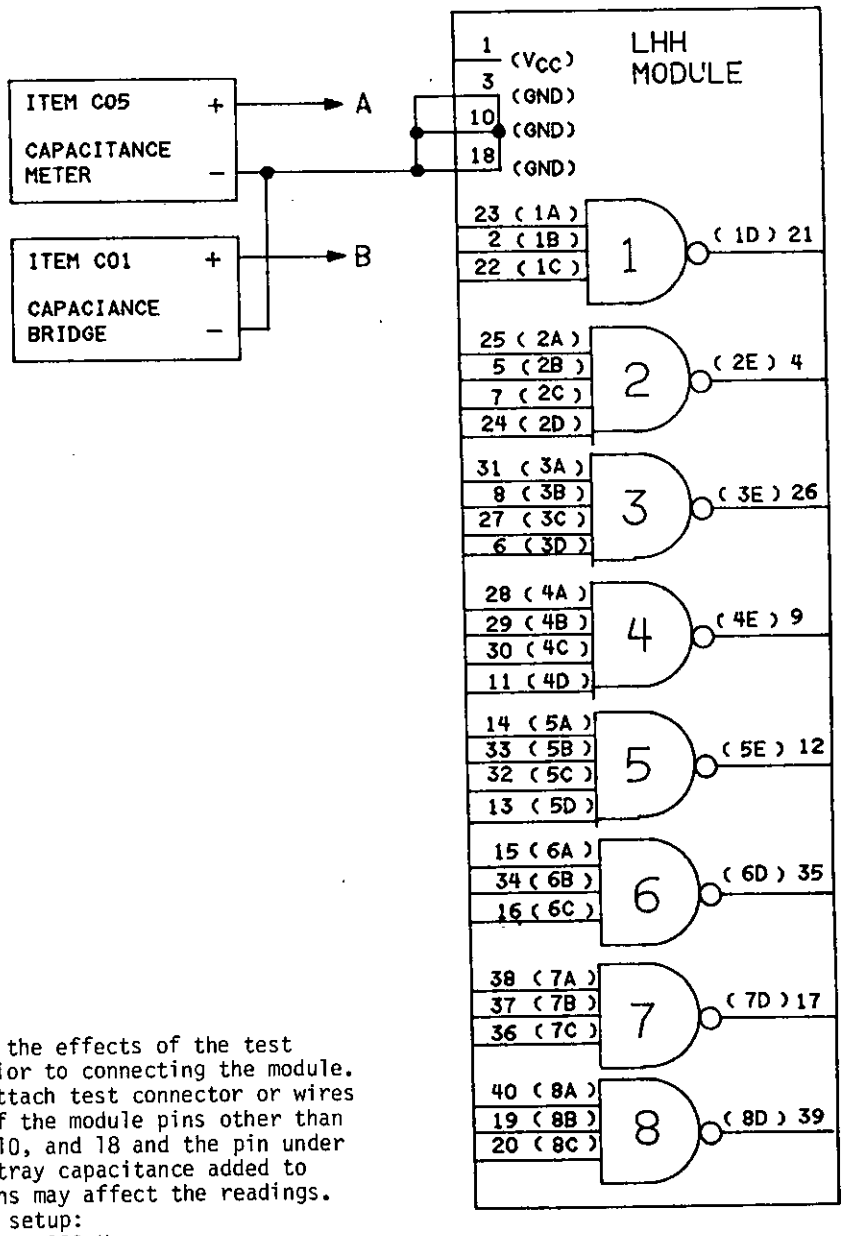
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NOTE: Input pulse parameters:
t_{TLH} = 3.5 ±1.5 ns (verified at each input pin).
PWA = 500 ±50 ns.
PRP = 1,000 ±100 ns.

FIGURE 24. Power supply transient amplitude test circuit waveforms for table XVIII.

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NOTES:
1. Null out the effects of the test leads prior to connecting the module.
2. Do not attach test connector or wires to any of the module pins other than pins 3, 10, and 18 and the pin under test. Stray capacitance added to other pins may affect the readings.
3. Item C01 setup:
   Frequency = 120 Hz.
   V_{pk} = 0.3 V maximum.

*FIGURE 25. Power supply filter capacitance and input capacitance test circuit for tables XIX and XX.

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(See Instructions – Reverse Side)

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3a. NAME OF SUBMITTING ORGANIZATION

b. ADDRESS *(Street, City, State, ZIP Code)*

4. TYPE OF ORGANIZATION *(Mark one)*

- ☐ VENDOR
- ☐ USER
- ☐ MANUFACTURER
- ☐ OTHER *(Specify)*: _____

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a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

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7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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